Imputing diffusion profiles of white matter fiber tracts in early brain development:

an extensive evaluation of normative, local, graph and transformer models

Martin A. Styner[1,2,3], Paul Friedrich[3], Aleeshah Nasir[1], Omar Azrak[1], Philippe C. Cattin[3], Heather C. Hazlett[1], John H. Gilmore[1], Weili Lin[4], Brittany R. Howell[5], Jed T. Elison[6], Jessica B. Girault[1]

[1]Department of Psychiatry, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

[2]Department of Computer Science, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

[3]Department of Biomedical Engineering, University of Basel, Allschwil, Switzerland

[4]Biomedical Research Imaging Center and Department of Radiology, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

[5]Fralin Biomedical Research Institute at VTC, Virginia Tech, Roanoke, VA, USA

[6]Institute of Child Development and Department of Pediatrics, University of Minnesota, Minneapolis, MN, USA

Corresponding author: Martin A. Styner (Department of Psychiatry and Department of Computer Science, University of North Carolina at Chapel Hill), styner@unc.edu

Abstract

Along-tract diffusion profiles are widely used to study white matter development, but in infant cohorts quality control rejects whole tracts in about half of the scans, and complete-case analyses lose these scans. We evaluated how such profiles can be imputed, with the `impute-profiles` command of the open-source DTIPlayground package, on 423 Baby Connectome Project scans (0–65 months; 69 tracts; FA, MD, RD, AD, NDI, ODI, FWF) and on a second infant cohort (HBCD). Measured values were hidden in five synthetic patterns and in the real missingness masks of other scans; 120 settings of seven method families and generic cohort imputers were compared on identical hidden values, with folds by subject, paired per-scan statistics and an untouched test set. (i) A scan's own neighbouring tracts do not predict a missing tract; imputation must be referenced to an age-appropriate norm. (ii) A linear population model of the deviations from the norm (probabilistic PCA over all tracts and measures) is a strong CPU-only baseline; for whole missing tracts, the dominant real pattern, it is best refined by k-nearest-neighbour imputation of its residuals. (iii) A transformer trained to correct the population model adds most where local information is scarce: blended with it, the error for whole profiles fell to 0.639 of the age-bin mean's (0.675 for population model + kNN), and for a missing tensor family from 0.696 to 0.424. On untouched subjects the blend's gain reproduced (paired ratio 0.925 against population model + SIREN; 0.916 for real masks). On HBCD the ranking transferred; the transformer alone matched or beat the blend with BCP weights for regions and real masks, and blend weights refitted on HBCD matched or beat it everywhere. (iv) A second cohort helped as training data for the transformer, but pooling it into the population model hurt; functa, a partial VAE and other more complex variants did not help. (v) At realistic missingness unrelated to group, imputation kept all scans with negligible bias in trajectory and group analyses, whereas complete-case analysis lost half of them; with missingness concentrated in one group, imputation shrank its group difference by about its imputed share (estimate 0.956 of the complete-data estimate). We give pattern-specific recommendations.

Keywords: diffusion MRI; tractometry; fiber tract profiles; missing data; imputation; normative modeling; infant brain development; transformer; probabilistic PCA

1 Introduction

Along-tract profiles sample diffusion measures such as fractional anisotropy (FA) or mean diffusivity (MD) at regular arc-length positions along a fiber tract, and compare those profiles across subjects (Goodlett et al., 2009; Colby et al., 2012; Yeatman et al., 2012; Verde et al., 2014; Chandio et al., 2020). They are widely used in studies of white matter development, where the measures change fast and non-linearly over the first years of life (Geng et al., 2012; Dubois et al., 2014; Lebel and Deoni, 2018; Ouyang et al., 2019; Stephens et al., 2020), and they increasingly include multi-compartment measures such as the neurite density (NDI), orientation dispersion (ODI) and isotropic free water fraction (FWF) of NODDI (Zhang et al., 2012; Dean et al., 2017). Infant and toddler data are, however, incomplete. Motion, susceptibility and signal dropout corrupt local regions, and quality control (QC) then rejects the tracts that pass through them; tracts are truncated where the tractography or the atlas mapping misses their ends; and some scans lack a model fit altogether. In the cohort studied here, 203 of 423 scans miss at least one profile value after QC, almost always whole tracts (Section 3.1). Many statistical tools for profiles (functional data analysis, mixed models over locations, multivariate and machine-learning analyses of all tracts jointly; Chamberland et al., 2019; Richie-Halford et al., 2021) assume complete profiles, so a scan that misses one tract is either discarded in full or patched ad hoc.

Imputation of missing values is a well studied problem (Rubin, 1976; Little and Rubin, 2019), with general methods ranging from nearest-neighbour (Troyanskaya et al., 2001), low-rank (Mazumder et al., 2010) and chained-equation (van Buuren and Groothuis-Oudshoorn, 2011) imputation to deep generative and masked models (Yoon et al., 2018; Mattei and Frellsen, 2019; Ma et al., 2019; Du et al., 2024). Tract profiles have structure that generic methods do not exploit: values vary smoothly along a tract, neighbouring tracts are dense in the brain, the measures of one scan are strongly coupled, and in a developmental cohort the expected value at each location depends strongly on age. It is not obvious which of these sources of information matters for which kind of missing value, how much a normative reference adds, and whether modern neural architectures bring anything beyond a well-specified linear population model.

This paper reports an extensive empirical evaluation of imputation methods for tract profiles in a normative infant and toddler cohort, the Baby Connectome Project (BCP; Howell et al., 2019), carried out while developing the `impute-profiles` command of the open-source DTIPlayground package (NIRAL, 2026). The contribution is not a new method but the knowledge the evaluation yields for practitioners. We organize the methods by the information they use, not by the order in which they were developed (the development history is in Appendix Table A1), and ask: (i) which approaches work for which missingness patterns (interior gaps, end gaps, whole profiles, all tracts through a region, a whole family of measures, and the real masks of the cohort); (ii) how much a normative reference helps; (iii) what linear population models, local three-dimensional (3-D) methods, generic cohort imputers, graph networks and transformers each contribute; (iv) what did not help; (v) whether and how a second, larger cohort (HBCD; Volkow et al., 2021; Dean et al., 2024) should be used; (vi) how the error varies by measure, age, tract and pattern; (vii) how well the methods do on subjects never used for any design decision and on a second cohort; and (viii) what imputation does to typical downstream analyses.

2 Related work

Tract profiles. Along-tract analysis was introduced for DTI by sampling tensor measures along atlas or subject fiber bundles (Goodlett et al., 2009; Colby et al., 2012) and popularized by automated pipelines such as AFQ (Yeatman et al., 2012), the UNC-Utah NA-MIC framework (Verde et al., 2014) and BUAN (Chandio et al., 2020), with tract segmentation by learned models in

recent pipelines (Wasserthal et al., 2018). Tractometry has been extended to many measures (Chamberland et al., 2019) and to multivariate analyses of all profiles of a scan (Richie-Halford et al., 2021); its reproducibility has been studied (Cousineau et al., 2017; Kruper et al., 2021). To our knowledge, the imputation of missing profile values has not been systematically evaluated.

Missing-data imputation. Classical results distinguish missing completely at random, at random and not at random (Rubin, 1976; Little and Rubin, 2019), and multiple imputation propagates the uncertainty of imputed values into downstream inference (Rubin, 1987). Practical imputers include k-nearest neighbours across samples (Troyanskaya et al., 2001), low-rank matrix completion (Mazumder et al., 2010), chained equations (van Buuren and Groothuis-Oudshoorn, 2011), random forests (Stekhoven and Bühlmann, 2012), and probabilistic PCA (PPCA) fitted by EM (Tipping and Bishop, 1999; Ilin and Raiko, 2010). Deep imputers include GAIN (Yoon et al., 2018), the partial VAE of EDDI (Ma et al., 2019), MIWAE (Mattei and Frellsen, 2019), graph-based GRAPE (You et al., 2020), masked-autoencoder ReMasker (Du et al., 2024) and the automatic model selection of HyperImpute (Jarrett et al., 2022); masked autoencoders (He et al., 2022) with transformers (Vaswani et al., 2017) are, in effect, amortized imputers. Benchmarks show that imputation quality and its effect on downstream models depend strongly on the data and the missingness (García-Laencina et al., 2010; Jäger et al., 2021; Shadbahr et al., 2023). In neuroimaging, imputation has been evaluated for regional morphometric measures (Sta Cruz et al., 2020) and for the imaging-derived phenotypes of large epidemiological cohorts (Radosavljević et al., 2025). Spatial interpolation by kriging or Gaussian processes (Cressie, 1993; Rasmussen and Williams, 2006) is the classical counterpart of our 3-D local methods.

Normative models. Normative modeling characterizes the distribution of a brain measure in a reference population as a function of covariates such as age, and expresses an individual as deviations from it (Marquand et al., 2016; Rutherford et al., 2022; Bethlehem et al., 2022), including in neonates (O'Muircheartaigh et al., 2020). Several of our methods impute such deviations instead of raw values.

Neural fields, graph networks and harmonization. Sine-activated networks (SIREN) represent signals as continuous functions of coordinates (Sitzmann et al., 2020); functa (Dupont et al., 2022) and MedFuncta (Friedrich et al., 2025) share one network across a dataset with per-instance latent modulations. Graph convolutional networks (Kipf and Welling, 2017) and graph U-Nets (Gao and Ji, 2019) operate on irregular structures; CONFETTI (Styner, 2026) builds a multi-resolution graph over tract-profile locations for scan classification. Multi-site pooling requires correcting site or acquisition effects, classically by location-and-scale models such as ComBat (Johnson et al., 2007; Fortin et al., 2017).

3 Materials and methods

3.1 Data

Cohorts. The normative cohort is the BCP (Howell et al., 2019): 423 scans of 208 children (Table 1, Fig. 1a). 105 children contributed two or more scans (up to 9). BCP ages are the nominal visit ages in whole months taken from the session label (`ses-012m`: 12 months), 0 to 65 months (median 13, quartiles 8 and 21). BCP ages are chronological (time since birth), not corrected for gestational age. Scans fall in three age bins used throughout: 0–3 months (28 scans), 4–9 months (117) and 10–60 months (278). The second cohort is HBCD (Volkow et al., 2021; Dean et al., 2024): 2,293 scans of 1,720 children, median age 1.4 months (range -1.5 to 15.8;

1,501, 658 and 133 scans in the three bins). The data are from HBCD release v2.0. The HBCD age is the candidate age at the MRI visit from the MRI RA prep instrument of that release (`mri_ra_prep_candidate_age`), in years, converted to months; negative values occur (minimum -1.5 months). HBCD ages are relative to full term (40 weeks of gestation), that is, term-corrected: a negative age means a scan before the term-equivalent date. HBCD scans were acquired at multiple sites on Siemens, GE and Philips scanners and preprocessed with QSIPrep (Cieslak et al., 2025) (Appendix A.1). The two cohorts were thus also preprocessed with different pipelines. The two cohorts therefore use different age scales: for an infant born before term, chronological age (BCP) exceeds term-corrected age (HBCD) by $40 - \mathrm{GA}$ weeks, GA being the gestational age at birth in weeks; for term-born infants the two agree within a few weeks. Gestational age is not available in our data. Different acquisitions and preprocessing pipelines (`dmriprep` for BCP, QSIPrep for HBCD; the model fitting and later steps were the same) contribute to the cohort offsets that the batch correction removes, and may leave differences that a shift and scale per cell do not capture. Within the bins the two cohorts differ in age composition (median 2.5 vs 1.0 months at 0–3 months, 18.0 vs 12.1 at 10–60 months).

Processing. After their respective preprocessing (BCP: `dmriprep`; HBCD: QSIPrep), both cohorts went through the same steps, run with DTIPlayground (model fitting with version 0.5.10; registration, profile extraction and QC with version 0.9.0) and the scripts that will be shared with the paper: the diffusion tensor (DIPY, weighted least squares, $b \leq 1600\,\mathrm{s/mm^2}$) and NODDI (AMICO, default settings) were fitted; each tensor image was registered to the age-appropriate mean tensor of a normative model of the DTI_IBISEP_Feb26 atlas (Short et al., 2022) (DTI-Reg, affine and greedy diffeomorphic ANTs registration) and checked by an automatic registration QC; and the profiles of the 69 atlas tracts were extracted in each scan's native space through its deformation field (details of every step in Appendix A.1). The normative model and the prior profile statistics of the QC were built from the full BCP cohort; they enter only the processing and QC, which defined the real missingness, and the norm of the released imputation default, never the validation, in which every norm and model was refitted without the validated subjects. *Profile QC* (`dmrifiberprofileqc-profiles`), which creates the real missingness: a scan was removed entirely if it failed the registration QC, if its preprocessing excluded more than 40% of its gradient volumes or more than 40% of the remaining volumes had an RMS motion above 2 mm (from the `dmriprep` QC report for BCP and the QSIPrep confounds for HBCD), or if it had no age; a tract of a scan (all its measures) was removed as a profile outlier if fewer than 80% of its positions, pooled over the measures except FWF, lay within 3.5 SD of the age-bin norm of a prior normative profile set after removing the cohort's median offset from that norm (or less than 75% of the profile was present), or if the correlation of its FA profile with the norm's mean profile was below $Q_1 - 3\,\mathrm{IQR}$ of its tract and age bin; locations sampled outside the brain mask were set missing on every measure of the tract, and diffusivities above $3.6 \times 10^{-3}\,\mathrm{mm^2/s}$ were removed as single values; finally, a scan missing more than half of its tracts was removed.

Profiles. Profiles were sampled on the 69 tracts of the DTI_IBISEP_Feb26 atlas, every 1 mm of arc length: 5,388 profile locations per measure. Each location has a 3-D position in atlas space, the point of the tract's axis (from the atlas fibers, `dmrifiberprofilecompute-axis`) at that arc length. Tract axes are dense: the median tract's locations are a median 2.4 mm from the nearest other tract's axis, and 37% of all locations are within 2 mm (76% within 4 mm) of another tract. Seven measures were analysed: FA, MD, radial and axial diffusivity (RD, AD) of the diffusion tensor, and NDI, ODI and FWF of NODDI (Zhang et al., 2012) (FWF is the isotropic fraction of the NODDI fit). The free-water tensor measures (Pasternak et al., 2009) were imputed in the

Table 1: The two cohorts (computed from the profile tables).

	BCP	HBCD
Scans	423	2,293
Subjects	208	1,720
Subjects with ≥ 2 scans	105	512
Scans per subject, median (max)	2 (9)	1 (3)
Age, months: median [IQR]	13.0 [8.0, 21.0]	1.4 [0.9, 5.8]
Age range, months	0.0–65.0	-1.5–15.8
Scans aged 0–3 / 4–9 / 10–60 mo	28 / 117 / 278	1,501 / 658 / 133
Scans outside the age bins	0	1
Tracts / profile locations	69 / 5,388	69 / 5,388
Measures	7	7
Cells missing after QC (%)	2.3	8.8
Scans with ≥ 1 rejected tract	201 (48%)	2,056 (90%)
Rejected tracts per affected scan, median (mean)	1 (3.3)	3 (6.8)
Scans missing $> 50\%$ of their tracts	2	98

development runs; they were dropped from the analyses because the free-water fits failed in many HBCD scans, so that both cohorts share the same measures; results with them for BCP are in Appendix Table A10. The cleaned profiles of the profile QC were used throughout.

Missingness of the real profiles. After QC, 2.29% of the BCP cells (365,899 of 15,953,868 scan × location × measure values) are missing, 99.8% of them in whole missing profiles: 4,599 of 204,309 scan × tract × measure profiles, always all measures of a tract at once (657 rejected scan × tract combinations). Only 168 profiles are partly missing (142 at their start or end, median 3 locations; 26 with an interior gap): 0.14% and 0.06% of the missing cells. 201 scans have at least one rejected tract; among them the median number of rejected tracts is 1 (75th percentile 3, 95th 12, maximum 63; 14 scans with 10 or more; Fig. 1b). 2 scans miss more than half of their tracts and are not imputed. Per tract, the rejection rate ranges up to 6.1% (AC_olfactory; median 2.1%) and is not related to the tract's proximity to other tracts (Spearman -0.17, $p = 0.15$; Fig. 1c). No BCP scan lacks a whole measure family after QC; that pattern arises in cohorts where a model fit fails or was not run, as the free-water fits of HBCD illustrate.

3.2 Notation and norms

For scan s, location p (tract, arc length) and measure m, y_{spm} is the profile value. The age-bin norm is the mean μ_{bpm} and SD σ_{bpm} of the cohort's scans in age bin b; the deviation of a scan is $z_{spm} = (y_{spm} - \mu_{b(s)pm})/\sigma_{b(s)pm}$. Methods "on the deviations" impute z and return $\hat{y} = \mu + \sigma\hat{z}$ with the norm of the missing location. A continuous norm replaces the bins by a local-linear kernel regression of each cell's mean on $\log(\text{age} + 1\,\text{month})$ (bandwidth by leave-one-out cross-validation). In validation, every norm is computed without the validated scan's subject.

3.3 Imputation methods

Table 2 summarizes the method families. All are implemented in `dtiplayground/dmri/fiberprofile/analysis/impute_profiles.py` (with `graph_unet.py`, `amortized.py` and `functa.py`); the generic cohort imputers are in the scripts of this paper.

A. Same-scan interpolation. *Arc-linear* interpolates linearly along the arc length between the observed values at both ends of a gap (up to 15 mm); missing ends are left, filled with the

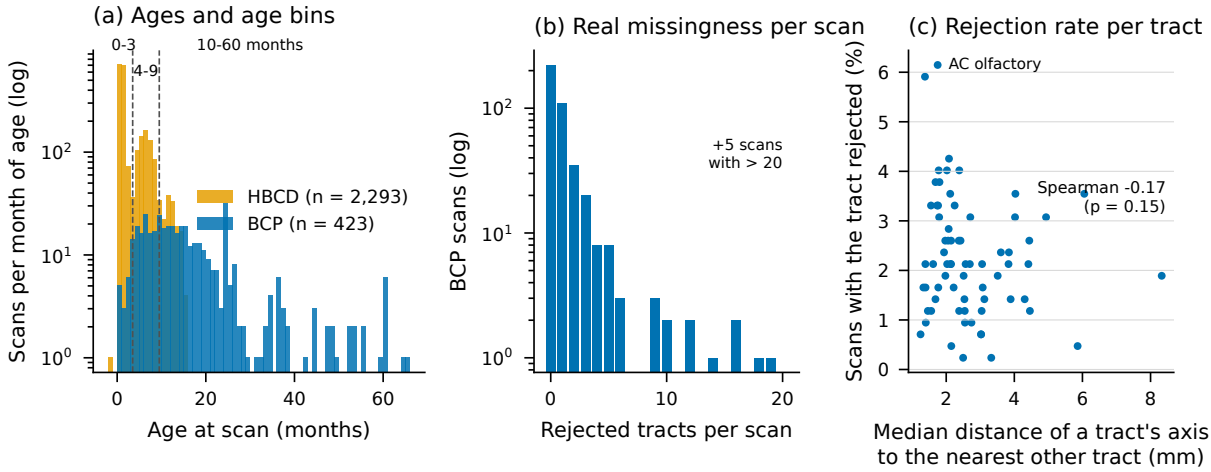


Figure 1: Data. (a) Ages at scan of BCP (normative cohort) and HBCD (second cohort); dashed lines: the age bins. (b) Real missingness of BCP after QC: number of rejected tracts per scan. (c) Per tract, the share of scans with the tract rejected against the median distance from the tract's axis to the nearest other tract.

nearest observed value (*constant*) or extrapolated from the three nearest. *Linear* interpolates barycentrically within the Delaunay tetrahedra of the observed 3-D positions of the same measure in all tracts. *kNN* averages the k nearest observed positions with inverse-distance weights. *SIREN* (Sitzmann et al., 2020) fits, per scan, a sine network (3 hidden layers of 256 units, Adam, learning rate 10^{-4}; Kingma and Ba, 2015) to all observed values of all measures jointly, each standardized within the scan, from the 3-D position (*xyz*), optionally with the arc length or a one-hot tract code; $\omega_0 \in \{5, 10, 20, 30\}$ and 100–1000 epochs were tested (Appendix Fig. A2).

B. Norm-referenced local methods. The same methods applied to z. Values farther than 20 mm from any observed value of the same measure are left to the age-bin mean.

C. Linear population model. The deviations of the normative scans over all locations and measures jointly ($5{,}388 \times 7$ cells observed in at least 10 scans) are modelled by PPCA (Tipping and Bishop, 1999), $\mathbf{z} = \boldsymbol{\mu} + W\mathbf{x} + \boldsymbol{\epsilon}$, $\mathbf{x} \sim \mathcal{N}(0, I_q)$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \psi I)$, fitted by EM with the exact posterior of every row (Ilin and Raiko, 2010), a ridge prior on the rows of W, and $q = 80$ components (5–160 tested; isotropic, per-measure and per-cell noise tested, Appendix Table A9). A scan's observed deviations give the posterior mean and SD of every missing cell, including all cells of a family the scan lacks. The posterior SDs are rescaled per measure so that errors on held-out whole profiles (5 folds by subject) have unit root-mean-square z. In *PPCA + local*, the PPCA prediction serves as the scan's own norm, and SIREN (xyz, $\omega_0 = 10$, 500 epochs) or kNN ($k = 20$) imputes the residuals within 4 mm of an observed value; beyond, the PPCA prediction is used. PPCA + SIREN was the default of DTIPlayground up to version 0.9.5.

D. Generic cohort imputers. To place PPCA among standard tools, we evaluated, on the same hidden values and the same training scans per fold: the *scan offset* (the norm plus the scan's mean observed deviation of the measure, or of the measure in the 5 tracts whose axis centroids are nearest to the missing tract); *kNN across scans* (Troyanskaya et al., 2001) (inverse-distance-weighted mean deviation of the $k \in \{5, 10, 20, 50\}$ training scans nearest in nan-Euclidean distance over the co-observed cells, as scikit-learn's `KNNImputer`); *iterative SVD* (hard-impute of rank $r \in \{10, 40, 80, 160\}$ on the training deviations, a scan's missing cells from a ridge projection of its

Table 2: Method families and what they use. Patterns: G interior gap, E end gap, W whole profile, R region, F whole measure family. Own: values of the scan itself; norm: age norm; cohort: models fitted on the normative cohort without the validated subject.

Method	Uses	Description	Patterns
A. Same-scan interpolation (the scan's own values)			
Arc-linear	own	linear along the arc length of the same profile; ends left, repeated or extrapolated	G, E
Linear (3-D), kNN, SIREN	own	Delaunay interpolation, inverse-distance kNN, or a per-scan sine network over the 3-D positions of all tracts	G, E, W, R
B. Norm-referenced local methods			
Any method of A on z	own, norm	imputes deviations from the age-bin norm	G, E, W, R
C. Linear population model			
PPCA	own, cohort	probabilistic PCA of the deviations of all locations and measures jointly	all
PPCA + kNN / SIREN	own, cohort	a local method imputes the PPCA residuals within 4 mm; PPCA beyond	all
D. Generic cohort imputers (deviation matrix, scans × 37,716 cells)			
Scan offset	own, norm	the scan's mean observed deviation (per measure, or of the 5 nearest tracts)	all
kNN across scans, iterative SVD, MICE	own, cohort	as `KNNImputer`, hard-impute with ridge projection, chained equations on tract means	all
E. Amortized networks (hybrid: correct the PPCA prediction)			
Graph U-Net	own, cohort	multi-resolution graph convolutional U-Net over the locations	all
Transformer	own, cohort	masked autoencoder over patches of 8 locations, global attention	all
F. Blends			
Fixed, precision, table	own, cohort	$(1-w)\,\hat{y}_{\text{PPCA+local}} + w\,\hat{y}_{\text{network}}$	all
G. Explored and not adopted			
Functa, partial VAE, age-dependent PPCA, acquisition correction	own, cohort	see text	all

observed cells on the loadings; cf. Mazumder et al., 2010); and *MICE* (van Buuren and Groothuis-Oudshoorn, 2011) (`IterativeImputer`, Bayesian ridge, 40 nearest features) on the tract-mean deviations (69×7 columns; MICE on 37,716 columns is impractical), whose imputed tract mean is assigned to every location of a hidden tract.

E. Amortized networks. Both networks receive per location the observed deviations (0 where missing), the observation mask, the 3-D position and the arc length, and, in the *hybrid* variant used throughout, the PPCA prediction and its log SD; they output a mean and log variance per location and measure and correct the PPCA prediction. They are trained on the normative cohort with the hiding patterns of Section 3.4 drawn anew in every epoch (a region in half of the scans, a whole family in a quarter), on the Gaussian negative log-likelihood of the hidden values (Kendall and Gal, 2017) (visible values with weight 0.1), with AdamW (Loshchilov and Hutter, 2019), a cosine schedule and batches of 16 scans; 10% of the training subjects select the epoch and calibrate the SDs, and the PPCA input of a training scan comes from a model fitted without its subject. The *graph U-Net*, after CONFETTI (Styner, 2026), uses a 4-level graph that keeps every 2^L-th location of each tract at level L, links nodes along the tract and to the closest node of each other tract within 1.5 times the median distance to the nearest other tract's axis, and a U-Net (Gao and Ji, 2019)

of graph convolutions (Kipf and Welling, 2017) (128 features, one convolution per level) with a scan-wide summary at the coarsest level; 60 epochs. CONFETTI's own configuration (graph from the dense fibers, 64 features, no summary) imputed poorly (Appendix Table A8). The *transformer* (Vaswani et al., 2017; He et al., 2022) cuts each tract into patches of 8 consecutive locations (about 700 tokens per scan); a token embeds the inputs of its patch plus a tract embedding; 6 pre-norm self-attention layers of width 192 (6 heads, dropout 0.1) relate every token to every other, and a linear head returns a mean and log variance for every location of the patch. It is trained for 200 or 400 epochs (learning rate 10^{-3}).

F. Blends. The PPCA + local prediction b and a network prediction g are combined as $(1-w)b + wg$, with (i) fixed weights w_{near} for cells within 4 mm (those the local method fills) and w_{beyond} for the others; (ii) precision weights $w = \sigma_b^{-2}/(\sigma_b^{-2} + \sigma_g^{-2})$; or (iii) a *weight table* per measure m and band k of the distance to the nearest observed value of the measure (0–0.5, 0.5–1, 1–2, 2–3, 3–4 mm for local cells; 4–6, 6–10, 10–20, >20 mm for fallback cells; and a measure the scan has no observed value of). With deviations in units of the age-bin SD, $t_i = (y_i - \mu_i)/\sigma_i$ the hidden truth and b_i, g_i the two predictions of hidden cell i, the table is the weighted least-squares solution

$$w_{mk} = \text{clip}_{[0,1]} \frac{\sum_{i \in (m,k)} c_i \, (t_i - b_i)(g_i - b_i)}{\sum_{i \in (m,k)} c_i \, (g_i - b_i)^2}, \tag{1}$$

where c_i is the inverse mean squared error of the age-bin mean in the cell's (pattern, measure, age bin) group, computed on the same training cells, so that each group counts as in the score (constrained stacking; Wolpert, 1992; Breiman, 1996). The table shipped with DTIPlayground 0.9.8 ("shipped") was fitted on all validation cells of seeds 0–2 with SIREN as the local method; scores on seeds 0–2 are cross-fitted (weights for a scan fitted without its subject, 5 folds). The SD of a blended value is $\sqrt{(1-w)\sigma_b^2 + w\sigma_g^2}$, where σ_b is the PPCA posterior SD (also for values filled by the local method). The released default of 0.9.8 is PPCA + SIREN blended with a 400-epoch transformer by the shipped table.

Second cohort. With `--transformer-reference-dir`, HBCD is batch-corrected to BCP and added to the training of the transformer only: the transformer is trained on the pooled scans with the deviations from their pooled age-bin norm and the pooled PPCA prediction as input, while the base prediction (norm, PPCA and SIREN) of a BCP scan remains that of BCP. The batch correction maps each cell to BCP by one shift and scale at the same age (a spline age curve with a shift per cohort, fitted jointly; the scale from the residual SDs over the common age range). Alternatively the cohorts can be pooled for the norm, the PPCA and the transformer.

G. Explored and not adopted. *Functa* (Dupont et al., 2022; Friedrich et al., 2025): a shared SIREN modulated per scan by shifts from a latent vector fitted by meta-learned gradient steps; we varied inner steps and step sizes, test-time refinement, an encoder start, depth (2–8), width (64–512), latent size (16–512), age and tract inputs, and graph variants, on the deviations and as a hybrid. *Partial VAE* (Ma et al., 2019; Mattei and Frellsen, 2019): a set encoder of the observed cells, a latent of 32–128, a per-location decoder. *Age-dependent PPCA*: a PPCA per age with kernel weights of log age. *Acquisition correction*: an offset per acquisition label and cell, fitted on the deviations with a random intercept per subject, removed before imputation. *Context inputs* to the transformer (distance to the nearest observed value, age and acquisition) and *seed ensembles* (three transformers averaged; Lakshminarayanan et al., 2017). The tuning budget per family is in Appendix Table A14.

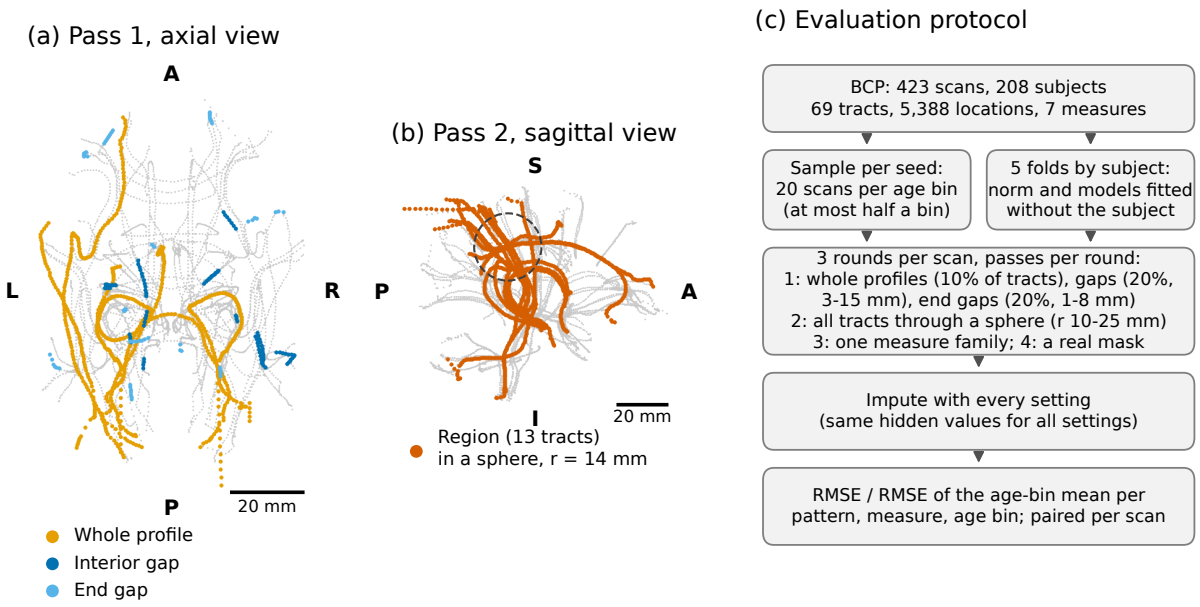


Figure 2: Missingness patterns and evaluation protocol. (a, b) One round on a 12-month BCP scan, the profile locations projected on axial and sagittal planes in atlas space (grey: observed; L/R, A/P, S/I: orientation; neurological convention). (a) Pass 1 hides whole profiles of random tracts, interior gaps and end gaps; (b) pass 2 hides all tracts through a sphere (dashed; this draw: radius as shown, 13 of 69 tracts, the median of the validation). Passes 3 (a measure family) and 4 (a real mask) not shown. (c) The protocol.

3.4 Evaluation protocol

Hiding patterns. Observed values are hidden, imputed and compared with the truth (Fig. 2). Validated scans must have at least 50% of each measure observed and at most 50% of their tracts missing; only observed values are hidden. (61 of the 131 scans validated in seeds 0–2 had real missing values.) Each validated scan goes through three rounds, each with up to four passes on the scan: (1) whole profiles (all measures) of 10% of the tracts with at least three observed locations; of the other tracts, an interior gap of 3–15 mm in 20% (*gap*) and the first or last 1–8 mm in another 20% (*end gap*); (2) every tract with an observed location inside a sphere of radius 10–25 mm around a random observed location loses its whole profile (*region*; spheres reaching more than half of the tracts are redrawn); (3) all values of one measure family the scan has, the tensor family (FA, MD, RD, AD) or the NODDI family (NDI, ODI, FWF) (*whole family*); and, in the runs of Sections 4.5–4.7, (4) the real missing cells of a randomly drawn donor, a BCP scan with missing values of the same age bin and another subject (*real mask*; `--validate-real-masks`; donor draws from their own random stream, so that passes 1–3 are unchanged). The hidden load of pass 1 (about 7 whole tracts and gaps in 40% of the other tracts) is much larger than that of the median real scan (1 rejected tract); the real-mask pass has the real load. The hiding is identical for all settings of a run and across runs with the same seed.

Sampling and folds. Per random seed (0, 1, 2), 20 scans per age bin were drawn, at most half of a bin (14, 20 and 20: 54 scans), 131 distinct scans of 96 subjects over the three seeds, 162 scan × seed pairs. Every model fitted on the cohort (norm, PPCA, continuous norms, batch correction, networks, blend weights, generic imputers) is fitted without the validated subjects: the validated scans are split by subject into 5 folds and each fold's models are fitted on all other subjects.

Untouched test (E1). All design choices (patterns, hyperparameters, blend variants, the shipped weight table) were made on seeds 0–2. For an unbiased estimate, three further draws (seeds 3–5) were restricted to subjects never validated in seeds 0–2 (`--validate-exclude-subjects`), with the shipped table scored as released. Only 168 eligible scans remain, so each draw validates 1, 16 and 20 scans in the three bins: the 0–3 month bin is essentially exhausted and E1 speaks for 4–60 months.

Second cohort as a test (E3). The same protocol on HBCD alone (seeds 0–2, 20 scans per bin), with all models fitted on HBCD folds, the transformer trained for 200 epochs (GPU budget for a cohort six times larger; 200 is also the validated setting with a reference cohort), the shipped BCP weight table, and a table refitted on HBCD (cross-fitted).

Pooling decomposition (E5). On the BCP scans of seeds 0–2, pooled with HBCD (batch-corrected to BCP): (i) pooled norm with a BCP-only PPCA (`--validate-model-batches`); (ii) BCP norm with a pooled PPCA, the HBCD deviations relative to the BCP norm (`--validate-norm-batches`); (iii) both pooled with the age-bin norm and with the kernel norm; compared with BCP alone on the same scans.

Score. For a setting, pattern P, measure m and age bin b, the RMSE over the hidden values is divided by the RMSE of the age-bin mean on the same values,

$$\rho_{Pmb} = \mathrm{RMSE}_{Pmb}(\text{setting})/\mathrm{RMSE}_{Pmb}(\text{age-bin mean}), \tag{2}$$

so that $\rho < 1$ is better than the norm. Values a setting leaves unfilled count with the age-bin mean. The pattern score is the mean of ρ over measures and age bins; the *overall* score is the mean over gap, end gap, whole profile and region. Because the overall score gives half its weight to gaps and end gaps, which are 0.14% + 0.06% of the real missing cells, we also report a *real-weighted* score (the pattern scores weighted by the real shares of missing cells, which is essentially the whole-profile score) and the real-mask pattern. As a reference-independent check we report the ratios relative to the leave-subject-out kernel age norm and absolute RMSEs (Appendix).

Paired statistics. The per-scan score is the mean over pattern × measure of $\sqrt{\mathrm{MSE}/\mathrm{MSE}_{\mathrm{norm}}}$ of that scan ($\mathrm{MSE}_{\mathrm{norm}}$: of the age-bin mean). Two settings A (reference) and B are compared by the geometric mean over scan × seed pairs of the ratio of their per-scan scores, with a 95% CI from a cluster bootstrap that resamples subjects (4,000 resamples; Field and Welsh, 2007), since seeds share scans and subjects share sessions; we also give the normal-theory CI over pairs, the Wilcoxon signed-rank test (Wilcoxon, 1945) on per-subject mean log ratios, Holm adjustment (Holm, 1979) over the 19 contrasts of Appendix Table A20 (selected after the development, not pre-specified), and the share of pairs in which B is better. For claims of "no difference" we use an equivalence margin of ±2%: a contrast is called equivalent when its bootstrap CI lies within 0.98–1.02 (Lakens, 2017). These intervals reflect the sampling of scans and hidden values, not the variation from network training (one network per fold and seed). Paired tests are only reported between settings that fill every hidden value.

Quick screens. Model families that did not justify a full validation (functa, partial VAE, age-dependent PPCA) and transformer variants were screened on one fixed held-out split (30 test subjects, two rounds, models fitted on the other subjects), scored like the validation. The best configuration of a family is selected on the test split itself, which favours the family; screen numbers are comparable within the screen only.

Downstream analyses (E4). To quantify what imputation does to typical analyses, we used the 423-scan cohort's 220 scans without any missing value (133 subjects, 5 folds by subject). For fold k, the age-bin norm and a transformer (0.9.8 settings, 400 epochs) were fitted on all BCP scans except those of fold k's subjects. The PPCA used for imputation is the transformer's hybrid-input PPCA ($q = 80$) saved with it: it was fitted on the transformer's training subjects of that fold cohort (about 90%; its 10% early-stopping holdout excluded), in deviations from the fold cohort's age-bin norm, without SD calibration. For multiple imputation its posterior SDs were scaled per measure by the factors of the population model of the default imputation run on the same fold cohort (all its scans, 5-fold calibration), which is a close but not identical model. In each of 50 replicates, a random half of the subjects formed group 1, and in 5 random tracts their values were shifted by δ between-scan SDs (all locations and measures; $\delta = 0.3$, and 0.1 in a second scenario); each scan then received the real missingness of a random donor (a BCP scan of the same age bin and another subject; about half of the donors have missing values, as in BCP; a third scenario uses only donors with missing values). Imputers: complete case (drop a scan with any missing value); available case (drop the scan for a tract with a missing value); age-bin mean; PPCA; PPCA + kNN; the blend of PPCA + kNN with the fold transformer using the shipped table (whose near-observed weights were fitted for SIREN cells); and multiple imputation with 10 draws from the calibrated PPCA posterior and Rubin's rules (Rubin, 1987). (SIREN per scan was too slow on the CPU for 50 replicates.) Analyses, per tract and measure (FA, MD) on the tract mean: (i) the trajectory, tract mean $\sim$ natural spline of $\log(\text{age} + 1)$ with 4 df, OLS with cluster-robust (subject) SEs, on the tracts without a planted effect, compared with the same fit on the complete data (integrated absolute bias in units of the tract mean's between-scan SD, CI width ratio, pointwise coverage of the complete-data curve); (ii) the group effect, tract mean $\sim$ group + spline, its estimate relative to the complete-data estimate, SE ratio, power and type I error.

Earlier protocols. The methods of families A–C were first run on 30 scans, one seed and 11 measures; all were re-run with the protocol above, and we report those runs throughout. The earlier runs are used only for the SIREN hyperparameter grid (Appendix Fig. A2), the noise models (Appendix Table A9) and the pooling with continuous norms on 11 measures (Appendix Table A11); the graph U-Net development runs are on seed 0 with 11 measures (Appendix Table A8).

3.5 Implementation and reproducibility

All methods are part of DTIPlayground 0.9.8 (NIRAL, 2026) (PyTorch, Paszke et al., 2019); the new validation options of this revision (`--validate-exclude-subjects`, `--validate-real-masks`, `--validate-norm-batches`, `--validate-model-batches`) are in DTIPlayground 0.9.9. Each validation run is one `dmrifiberprofileimpute-profiles--validate` call; with `--validate-dump-dir` it also writes every hidden cell with all predictions, from which the blends, the weight tables, the generic imputers and the calibration were scored offline. A run with the population model, SIREN and one network took about 2 hours on one NVIDIA RTX A6000 GPU. Every number, table and figure is produced from the result files by the scripts accompanying the paper.

4 Results

4.1 Overview

Table 3 and Fig. 3 give the error of representative settings on the same hidden values; Table 4 adds the real-weighted score and the reference-independent view.

Table 3: Error relative to the age-bin mean (RMSE ratio ρ; below 1: better) on identical hidden values, mean over 3 seeds × 54 BCP scans, 7 measures and 3 age bins. Overall: mean of the four patterns (SD between seeds). †Leaves some hidden values, counted with the age-bin mean (families A–B never fill a whole family). *Weights cross-fitted by subject; mean of the per-seed scores (pooled over seeds: 0.567 and 0.538).

		Pattern				Whole family missing	
Setting	Overall (SD)	Gap	End	Whole	Region	Tensor	NODDI
Reference							
Age-bin mean	1.000 (0.000)	1.000	1.000	1.000	1.000	1.000	1.000
A. Same-scan interpolation (the scan's own values)							
Arc-linear, constant ends†	0.900 (0.015)	0.759	0.842	1.000	1.000	1.000	1.000
Linear (3-D Delaunay)†	1.496 (0.043)	1.074	0.968	1.818	2.123	1.000	1.000
kNN, $k = 20$†	1.713 (0.048)	1.490	1.198	1.893	2.269	1.000	1.000
SIREN, xyz†	1.418 (0.045)	0.780	0.823	1.831	2.239	1.000	1.000
SIREN, xyz+tract†	1.572 (0.057)	0.732	0.812	2.335	2.408	1.000	1.000
B. Norm-referenced local methods (deviations from the age-bin norm)							
Arc-linear, constant ends†	0.750 (0.011)	0.431	0.570	1.000	1.000	1.000	1.000
Linear (3-D Delaunay)†	0.733 (0.006)	0.496	0.701	0.801	0.934	1.000	1.000
kNN, $k = 20$†	0.755 (0.009)	0.631	0.683	0.782	0.923	1.000	1.000
kNN, $k = 50$†	0.759 (0.010)	0.663	0.712	0.767	0.893	1.000	1.000
SIREN, xyz+arc†	0.704 (0.005)	0.422	0.516	0.895	0.982	1.000	1.000
SIREN, xyz†	0.688 (0.004)	0.442	0.544	0.814	0.952	1.000	1.000
C. Linear population model (PPCA, $q = 80$)							
PPCA alone	0.733 (0.008)	0.685	0.777	0.713	0.759	0.696	0.798
PPCA + kNN on residuals (≤4 mm)	0.664 (0.008)	0.568	0.659	0.675	0.752	0.696	0.798
PPCA + SIREN on residuals (≤4 mm)	0.614 (0.007)	0.440	0.552	0.697	0.766	0.696	0.798
E. Amortized networks and F. blends: graph U-Net (hybrid)							
U-Net alone	0.673 (0.008)	0.596	0.686	0.676	0.735	0.530	0.703
U-Net + SIREN on residuals	0.659 (0.014)	0.485	0.629	0.741	0.780	0.530	0.703
PPCA + SIREN, U-Net beyond 4 mm	0.610 (0.007)	0.436	0.550	0.697	0.757	0.530	0.703
E/F. Transformer (hybrid) and blends							
Transformer alone, 200 epochs	0.654 (0.009)	0.547	0.627	0.704	0.738	0.446	0.636
Transformer alone, 400 epochs	0.632 (0.005)	0.510	0.588	0.694	0.737	0.425	0.623
PPCA + SIREN, transformer beyond 4 mm	0.606 (0.008)	0.438	0.542	0.692	0.750	0.446	0.636
Blend with U-Net, fixed 0.25/0.75	0.594 (0.006)	0.435	0.541	0.665	0.736	0.540	0.705
Blend with transformer (200), fixed 0.25/0.75	0.582 (0.008)	0.424	0.518	0.658	0.729	0.460	0.643
Blend with transformer (400), fixed 0.25/0.75	0.576 (0.006)	0.414	0.509	0.654	0.726	0.443	0.630
Blend with transformer (400), weight table*	0.569 (0.005)	0.418	0.505	0.639	0.715	0.424	0.622
E/F. Transformer (200 epochs) trained with the HBCD reference cohort							
Reference transformer alone	0.580 (0.013)	0.456	0.532	0.633	0.698	0.432	0.662
Blend, fixed 0.25/0.75	0.557 (0.009)	0.400	0.495	0.634	0.697	0.439	0.642
Blend, weight table*	0.540 (0.010)	0.398	0.481	0.603	0.678	0.429	0.645

Table 4: Main settings by the patterns that dominate real missingness: whole profiles, regions and the real-weighted score (pattern scores weighted by the real shares of missing cells), relative to the age-bin mean; and relative to a leave-subject-out kernel age norm (overall, whole profiles, regions). Seeds 0–2.

	vs age-bin mean			vs kernel age norm		
Setting	Whole	Region	Real-weighted	Overall	Whole	Region
Age-bin mean	1.000	1.000	1.000	1.100	1.108	1.087
Kernel age norm (alone)	0.912	0.927	0.912	1.000	1.000	1.000
Phase 2: SIREN on deviations	0.814	0.952	0.814	0.749	0.893	1.028
PPCA alone	0.713	0.759	0.713	0.799	0.783	0.819
PPCA + kNN	0.675	0.752	0.675	0.722	0.740	0.811
PPCA + SIREN	0.697	0.766	0.697	0.668	0.765	0.826
Graph U-Net alone	0.676	0.735	0.676	0.732	0.742	0.793
Transformer alone (400)	0.694	0.737	0.693	0.688	0.761	0.795
Blend, fixed (400)	0.654	0.726	0.654	0.626	0.718	0.783
Released default (table)	0.639	0.715	0.639	0.619	0.701	0.771
HBCD-trained transformer, table	0.603	0.678	0.603	0.587	0.661	0.731

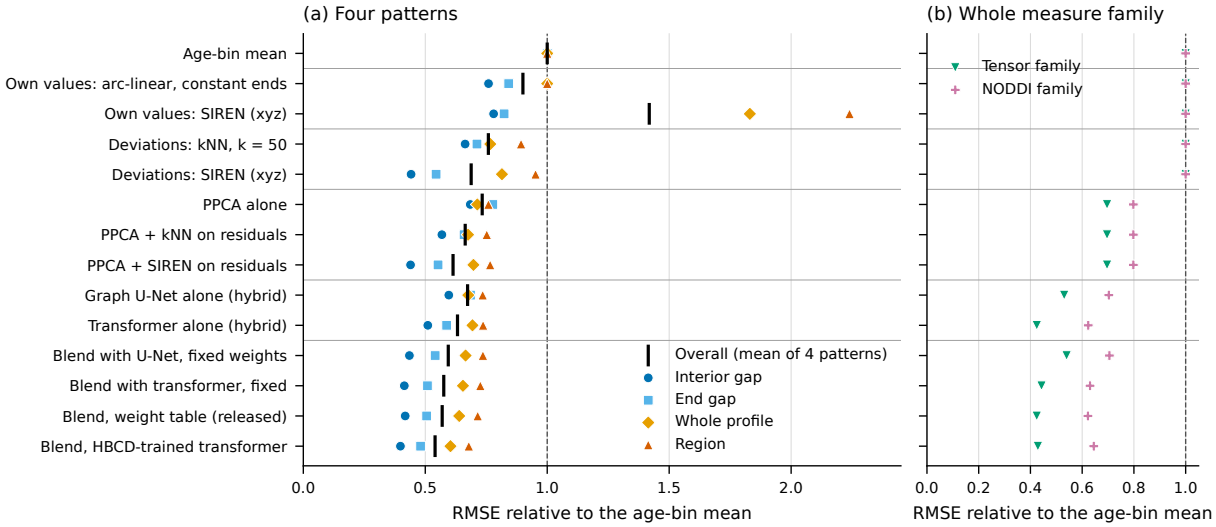


Figure 3: Error relative to the age-bin mean of representative settings, grouped by method family. (a) Per pattern (markers) and overall (vertical bar). (b) A whole measure family hidden; same-scan and norm-referenced local methods cannot impute it (1.0 by construction). "Released": the default of DTIPlayground 0.9.8 (cross-fitted).

Same-scan interpolation is not enough. From a scan's own values, only gaps and end gaps along a profile were imputed better than the age-bin mean (arc-linear 0.759 and 0.842 with constant ends; SIREN 0.780 and 0.823). The 3-D methods imputed whole profiles and regions far worse than the norm (SIREN 1.831 and 2.239, linear 1.818 and 2.123): a scan's neighbouring tracts do not predict the values of a missing tract.

The norm makes every pattern imputable. On the deviations from the age-bin norm, SIREN reached 0.814 for whole profiles and 0.952 for regions (from 1.831 and 2.239 on the values), and 0.688 overall. The norm carries the tract-specific level at the scan's age; the neighbours then only need to carry the scan's deviation from it. Regions remained close to the norm.

A linear population model. PPCA alone improved whole profiles (0.713) and regions (0.759) over the norm-referenced local methods, and is the first method that imputes a missing measure family (tensor 0.696, NODDI 0.798). Imputing its residuals locally helped gaps and end gaps most.

Table 5: Pattern-specific paired contrasts (geometric mean ratio of per-scan errors, compared / reference, with the 95% subject-bootstrap CI; 162 scan × seed pairs).

Contrast (compared vs reference)	Whole profile	Region	Interior gap	End gap
PPCA + kNN vs PPCA + SIREN	0.965 [0.960, 0.969]	0.981 [0.978, 0.983]	1.325 [1.303, 1.348]	1.187 [1.163, 1.211]
PPCA alone vs PPCA + SIREN	1.026 [1.017, 1.036]	0.990 [0.986, 0.994]	1.608 [1.576, 1.642]	1.406 [1.368, 1.441]
Transformer (400) vs PPCA + SIREN	0.994 [0.983, 1.006]	0.955 [0.947, 0.963]	1.185 [1.160, 1.210]	1.026 [0.995, 1.058]
U-Net vs PPCA + SIREN	0.963 [0.953, 0.974]	0.956 [0.948, 0.963]	1.370 [1.341, 1.403]	1.217 [1.187, 1.246]
Released default vs PPCA + SIREN	0.915 [0.908, 0.922]	0.929 [0.923, 0.934]	0.952 [0.940, 0.964]	0.907 [0.892, 0.922]
Released default vs PPCA + kNN	0.949 [0.942, 0.956]	0.947 [0.941, 0.953]	0.718 [0.705, 0.732]	0.764 [0.750, 0.778]
Released default vs PPCA alone	0.892 [0.884, 0.900]	0.939 [0.933, 0.944]	0.592 [0.581, 0.604]	0.645 [0.631, 0.660]
HBCD-reference blend vs released default	0.950 [0.939, 0.960]	0.956 [0.945, 0.966]	0.954 [0.944, 0.965]	0.943 [0.932, 0.956]

Which local method is better depends on the pattern (Table 5): for gaps SIREN was much better than kNN (kNN 1.325 times SIREN's error), but for whole profiles kNN was better (0.965, 95% CI 0.960–0.969, better on 90% of scan draws) and for regions slightly better (0.981); for regions PPCA alone was as good as PPCA + SIREN (0.990). Under the real weighting, PPCA + kNN (0.675) is better than PPCA + SIREN (0.697).

Distance limit of the local methods. The PPCA residuals are local. Scored from the per-distance errors of the same cells (Appendix Table A3), SIREN on the residuals up to 2 mm instead of 4 mm improved whole profiles (0.659 vs 0.665) and regions (0.722 vs 0.731) but made gaps worse (0.486 vs 0.416); for kNN the 4-mm limit was better for every pattern. The 4-mm limit of SIREN is therefore a compromise driven by the gap patterns.

Generic cohort imputers. Standard imputers on the deviation matrix, with the same training scans as PPCA (Appendix Table A2), were clearly worse than the population model, except for the low-rank model, which is essentially the same model fitted differently: iterative SVD of rank 160 reached 0.727 overall and 0.709 for whole profiles (paired against PPCA alone for whole profiles 0.990, CI 0.979–1.000). kNN across scans (best $k = 20$: 0.837 overall, 0.821 for whole profiles; paired 1.134) and MICE on the tract-mean deviations (0.843; 0.820; paired 1.146) were about 13–15% worse than PPCA for whole profiles. The scan-wide offset alone (the norm plus the scan's mean observed deviation of the measure) reached 0.887 for whole profiles (0.876 using the 5 nearest tracts), that is, a scan-wide shift accounts for about 39% of PPCA's gain over the norm on whole profiles; the rest is the covariation of deviations across tracts and measures.

Amortized networks and blends. The hybrid graph U-Net (0.673) and transformer (0.632 with 400 epochs) alone were worse than PPCA + SIREN overall, because they are worse for gaps and end gaps; for whole profiles the U-Net was better than PPCA + SIREN (paired 0.963, CI 0.953–0.974) and the transformer equivalent (0.994, CI 0.983–1.006), and both were better for regions (0.956, 0.955; Table 5). Their strength is a missing family: tensor 0.530 and 0.425, NODDI 0.703 and 0.623, against 0.696 and 0.798 for PPCA (paired, transformer vs PPCA for a missing family: 0.667, better on 100% of draws). Blending the networks with PPCA + SIREN combined both: a fixed blend reached 0.594 with the U-Net and 0.576 with the transformer, the weight table 0.569 (cross-fitted). Against PPCA + SIREN, the released default had a paired ratio of 0.915 for whole profiles, 0.929 for regions, 0.952 for gaps and 0.907 for end gaps (better on 98%, 100%, 73% and 85% of draws), and against PPCA + kNN 0.949 for whole profiles (Table 5). The transformer was better than the U-Net alone (paired 0.976), but the U-Net had a smaller training budget (60 epochs; Appendix Table A14). Trained for 200 epochs on the same draws (O5), the U-Net did not improve: alone it reached 0.692 (paired against 60 epochs 1.026, CI 1.022–1.030; the epoch with the smallest held-out error is kept, so longer training only adds overfitting room),

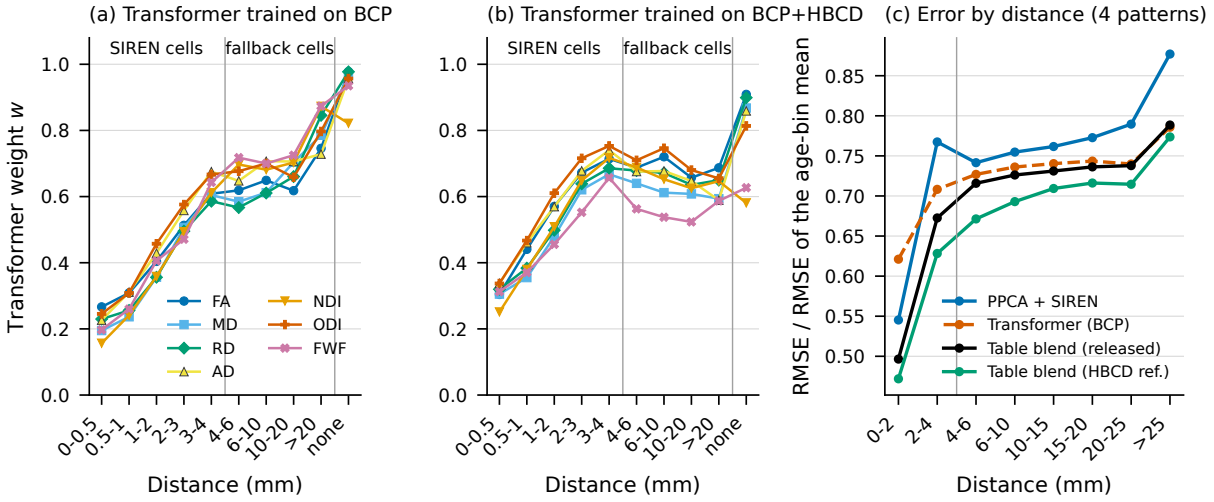


Figure 4: Blend weights. (a, b) Weight w of the transformer per measure and distance band (local cells within 4 mm, fallback cells beyond, "none": no observed value of the measure), fitted on all validation cells; (a) trained on BCP (shipped table), (b) trained on BCP and HBCD. (c) Error against distance, four patterns pooled, from the cell dumps (distance bins as in Fig. 5; PPCA alone is in Fig. 5).

its fixed blend 0.595 (60 epochs: 0.594), and against the transformer at the same 200 epochs its error was 1.050 times as large (CI 1.042–1.058). The advantage of the transformer is therefore not a training-budget artefact.

Blend weights. The learned weight of the transformer rose with the distance to the nearest observed value (Fig. 4): 0.16–0.27 next to an observed value, 0.57–0.73 for fallback cells, 0.82–0.98 for a missing family. Refitting reproduced the shipped table to two decimals. The table was better than fixed 0.25/0.75 weights with the same transformer (0.987, CI 0.984–0.989), 400 epochs better than 200 (0.988, CI 0.986–0.990); precision weights were worse (1.012, CI 1.010–1.015), because the SD attached to a locally imputed value is that of PPCA, an upper bound next to an observed value. A gradient-boosted weight per cell and calibrated precision weights were no better than the table (Appendix Table A7). These differences of 1–2% were selected on the same draws (Section 4.5).

Distance to the nearest observed value. Within each pattern, the error grew with the distance to the nearest observed value of the same measure (Fig. 5). PPCA + SIREN was worse than PPCA alone at 2–4 mm for whole profiles and regions, where PPCA + kNN was not; beyond 4 mm the networks and the blend were best.

4.2 How much does the normative reference help?

The normative information enters at three levels. The age-bin norm turned whole profiles and regions from 1.8–2.4 times the norm's error (same-scan 3-D methods) into 0.77–0.95 (norm-referenced local methods, Table 3). The population model, which captures how deviations co-vary across all tracts and measures, took whole profiles to 0.713 and made a missing family imputable. Learned corrections of the population prediction took whole profiles to 0.639 (0.603 with a larger training cohort). A continuous kernel age norm alone was 0.912 of the age-bin mean's error for whole profiles, and the age-bin mean was 1.000 / 0.912 ≈ 1.10 of the kernel norm's (Table 4); relative to the kernel norm all ratios rise by about 10% but the ranking is unchanged (whole profiles: released default 0.701, PPCA + kNN 0.740, PPCA + SIREN 0.765). Under PPCA + SIREN the choice

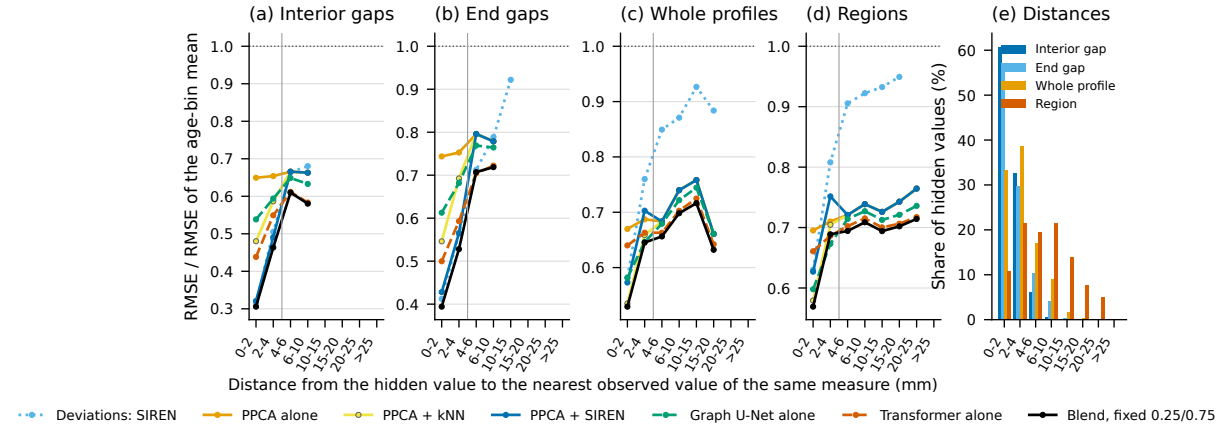


Figure 5: Error by the distance from a hidden value to the nearest observed value of the same measure, per pattern (3 seeds; bins with fewer than 600 hidden values omitted; vertical line: 4 mm). (e) Share of hidden values per distance bin.

of norm made no difference (kernel vs bins 1.005, CI 0.996–1.015, 11-measure protocol, Appendix Table A11): the population model absorbs the within-bin age trend.

4.3 Size of the normative cohort and age resolution of the norm

To see how many normative scans the population model needs (O1), the norm and the PPCA of each validation fold were fitted on a random 25, 50, 75 or 100% of the training subjects (on average 99, 198, 295 and 395 scans), and PPCA alone and PPCA + kNN were scored on the same hidden values (Appendix Table A16, Fig. A4; CPU fits, 3 seeds, 162 scan × seed pairs). For whole profiles, PPCA + kNN reached 0.857, 0.734, 0.691 and 0.675 with the four training sets; paired against the full set, the error was 1.187 (CI 1.151–1.226) times as large with about 99 scans, 1.073 (CI 1.056–1.094) with about 198 and 1.022 (CI 1.013–1.031) with about 295. The gain thus flattens above about 300 scans but has not stopped at 400, so a larger normative cohort would still help a little. Because the norm was refitted from the same subsets, the curve includes the effect of a noisier age-bin norm.

Finer age bins (O3; seven bins of 0–1, 2–3, 4–6, 7–9, 10–14, 15–24 and 25–65 months) made the norm alone a better predictor (0.951 of the 3-bin mean's error overall), but under the population model they were worse than the three bins (PPCA + kNN, whole profiles: 0.708 against 0.675; paired 1.035, CI 1.020–1.051): with fewer scans per bin, the norm's means and SDs are noisier, and the population model already absorbs the within-bin age trend (as with the kernel norm above).

4.4 What did not help

Functa. Over 187 screened runs, the best functa model reached 0.702 as a hybrid correcting PPCA and 0.736 on the deviations, against 0.720 for PPCA alone and 0.654 for the hybrid U-Net on the same split (Appendix Table A17, Fig. A1). The hybrid functa is thus 2.5% better than PPCA alone on the screen (optimistically, being selected on that split), but half as much as the U-Net gained; it was never combined with SIREN or blended. Fitting a per-scan latent by a few gradient steps did much worse than PPCA's exact posterior, even with a learned vector per location that could in principle reproduce PPCA.

Partial VAE and age-dependent PPCA. The partial VAE reached 0.730 as a hybrid and 0.807 on the deviations (4 configurations each, a small budget), no better than PPCA alone. A

Table 6: E1: untouched subjects (seeds 3–5; subjects of seeds 0–2 excluded; 1, 16 and 20 scans per draw in the three bins), the shipped weight table scored as released. "Real mask": the fourth pass.

Setting	Overall (SD)	Gap	End	Whole	Region	Tensor	NODDI	Real mask
age-bin mean	1.000 (0.000)	1.000	1.000	1.000	1.000	1.000	1.000	1.000
PPCA alone	0.803 (0.016)	0.748	0.799	0.811	0.853	0.760	0.859	0.801
PPCA + kNN	0.731 (0.020)	0.615	0.700	0.763	0.844	0.760	0.859	0.758
PPCA + SIREN	0.680 (0.015)	0.470	0.603	0.787	0.861	0.760	0.859	0.777
Transformer alone	0.699 (0.013)	0.569	0.644	0.778	0.803	0.480	0.633	0.757
Fixed blend (SIREN base)	0.639 (0.011)	0.447	0.569	0.736	0.803	0.491	0.651	0.723
Fixed blend (kNN base)	0.682 (0.014)	0.558	0.647	0.729	0.795	0.491	0.651	0.718
Precision blend	0.643 (0.015)	0.476	0.565	0.734	0.796	0.507	0.679	0.724
Shipped table (SIREN base; released default)	0.633 (0.012)	0.456	0.563	0.723	0.789	0.479	0.636	0.713
Shipped table (kNN base)	0.670 (0.015)	0.541	0.625	0.725	0.787	0.479	0.636	0.716

population model per age with kernel weights was worse than one model of all ages (0.732–0.755 against 0.720).

Acquisition correction. BCP scans of the `shells06select` acquisition deviated systematically from `dir79select` (FA +0.41, AD +0.47, NDI +0.52, ODI -0.26 SD, fitted with a random intercept per subject), and the two acquisitions are unevenly distributed over the age bins (Appendix Table A13). Removing these offsets before imputation was equivalent to not removing them (paired 1.001, CI 0.999–1.004). A scan-wide offset is visible in the scan's own observed values; how much of PPCA's gain is such an offset is shown by the scan-offset baseline above.

Context inputs and seed ensembles. On seed 0 (54 scans), giving the transformer the distance to the nearest observed value (blend, paired 1.001, CI 0.999–1.002) or the age and acquisition (1.000, 0.998–1.002), or averaging three transformers trained with different seeds (0.999, 0.998–1.000), was equivalent within ±2% to the plain transformer. These checks used one seed.

4.5 Untouched test subjects

On subjects never used for any design decision (Table 6; 111 scan × seed pairs of 63 subjects), the conclusions held. The released default had a paired ratio of 0.925 (CI 0.919–0.931, better on 98% of draws) against PPCA + SIREN, against 0.923 on the development draws; for whole profiles 0.915 (CI 0.908–0.922; development 0.915). The ranking for whole profiles was unchanged: released default 0.723, PPCA + kNN 0.763, PPCA + SIREN 0.787, PPCA alone 0.811 (PPCA + kNN vs PPCA + SIREN 0.965, CI 0.958–0.971; development 0.965), and SIREN remained much better for gaps (1.324 for kNN). The shipped weight table was again slightly better than fixed weights (0.988, CI 0.985–0.991). Blending the transformer with PPCA + kNN instead of PPCA + SIREN, with the same table, gave 0.915 of PPCA + kNN's error (CI 0.910–0.920). All ratios to the age-bin mean were higher than on the development draws (released default 0.633 overall, 0.586 at 4–60 months; PPCA + SIREN 0.680 and 0.631); the untouched scans are those left after the development draws and lie farther from their bin means, but the relative gains between methods, which the paired ratios measure, reproduced within 0.002. The 0–3 month bin had only 1 scan per draw, so E1 speaks for 4–60 months. A weight table refitted on the untouched cells (cross-fitted) gave the same error as the shipped table (0.633 vs 0.633), so the shipped weights are not overfitted to the development draws. Relative to the kernel age norm, which on these scans was 0.966 of the age-bin mean's error, the released default reached 0.654 overall and 0.753 for whole profiles (PPCA + kNN 0.795, PPCA + SIREN 0.821). Its SDs were calibrated as on the development draws (RMS error/SD 0.86 for whole profiles, 97% within 1.96 SD; 0.88 and 97% for real masks).

Table 7: E2: error relative to the age-bin mean for the real-mask pattern (the real missing cells of a donor scan of the same age bin and another subject), with all donors and with the donors that `impute-profiles` would impute (at most half of their tracts missing). Cell dumps of E1 (seeds 3–5) and E3 (seeds 0–2); PPCA alone and the pooled runs are not in the dumps (all donors: Appendix Table A22).

	BCP, untouched subjects (E1)		HBCD (E3)	
Setting	all donors	donors ≤50% tracts missing	all donors	donors ≤50% tracts missing
Age-bin mean	1.000	1.000	1.000	1.000
PPCA + kNN	0.758	0.752	1.083	0.973
PPCA + SIREN	0.777	0.771	1.086	0.975
Transformer alone	0.757	0.754	0.767	0.743
Fixed blend (SIREN base)	0.723	0.718	0.816	0.786
Shipped table (released default)	0.713	0.708	0.799	0.768
Shipped table (kNN base)	0.716	0.712	0.805	0.775
Table refitted on the cohort (cross-fitted)	0.712	0.707	0.764	0.737

4.6 Realistic missingness

With the real missing cells of other scans as the hidden values (Table 7, Appendix Table A22; untouched subjects), the errors were close to those of whole profiles, as expected from the real missingness. Unlike the per-scan paired ratios, this score is cell-weighted: donors with many rejected tracts contribute most of its cells. The validation draws donors from every scan with missing values, including the few that miss more than half of their tracts and would not be imputed by `impute-profiles` (`--max-missing-tracts` 0.5); we recovered the donor of every pass (by replaying the validation's random draws, verified against the hidden cells) and report both all donors and donors within the tool's scope. For BCP the restriction hardly matters (2 of 325 passes; released default 0.708 instead of 0.713). With all donors, the released default had 0.713 of the age-bin mean's error, PPCA + kNN 0.758, PPCA + SIREN 0.777 and PPCA alone 0.801. Paired, the released default was 0.916 of PPCA + SIREN's error (CI 0.905–0.927, better on 94% of draws) and 0.948 of PPCA + kNN's (CI 0.936–0.960); PPCA + kNN was 0.966 of PPCA + SIREN's (CI 0.955–0.977), and PPCA alone 1.032 (CI 1.014–1.049). The realistic masks thus confirm the ranking derived from the whole-profile pattern: released default, then PPCA + kNN, then PPCA + SIREN. In BCP the error depended little on how much of the scan was hidden (Fig. 6a; Appendix Table A23). In the development results, weighting the four patterns by the real frequencies amounts to the whole-profile score (Table 4); weighting the whole-profile error of each tract by its real rejection rate changed the released default's error from 0.583 (unweighted mean over tracts) to 0.589, since the per-tract error is unrelated to the rejection rate (Spearman 0.07, $p = 0.59$). In regions, the error grew with the number of tracts hidden together (released default 0.658 for 1–5 tracts to 0.724 for more than 20; PPCA + SIREN 0.690 to 0.776).

4.7 A second cohort

HBCD as a test cohort (E3). On HBCD alone (Table 8; 180 scan × seed pairs of 159 subjects; every model fitted on HBCD folds, the transformer trained for 200 epochs, the shipped BCP weight table), the main conclusions transferred, with two differences. The released default had 0.872 of PPCA + SIREN's error (CI 0.862–0.881; BCP 0.923), 0.898 for whole profiles and 0.828 for the real masks of HBCD (CI 0.808–0.848). PPCA + kNN was again slightly better than PPCA + SIREN for whole profiles (0.982, CI 0.974–0.989) and real masks (0.990) and much worse for gaps (1.399), and the shipped table again beat fixed weights (0.983). The differences: first, the population model gained little for regions (0.978 for PPCA + SIREN, 0.970 for PPCA + kNN) and for the real masks of HBCD. HBCD's missingness is heavier and more clustered than BCP's (Table 1: 8.8% of the cells missing vs 2.3%, median 3 rejected tracts per affected scan vs 1), and its real-mask passes hid about twice as much (median 204 locations vs 98 in BCP; mean 513 vs 231). 20 of 433 HBCD

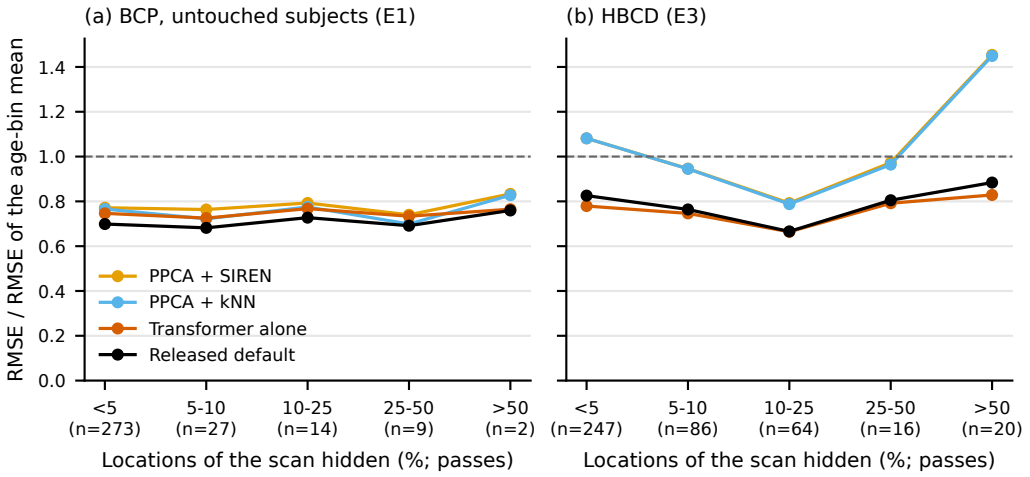


Figure 6: Real-mask pattern: error relative to the age-bin mean against the share of the scan's 5,388 locations hidden (passes per class in parentheses). Above 50%, nearly all donors are outside the scope of `impute-profiles`. (a) BCP, untouched subjects (E1); (b) HBCD (E3).

passes (4.6%) used a donor outside the scope of `impute-profiles`, hiding up to 97% of the scan, and they held 36% of the real-mask cells. With all donors, PPCA + SIREN was worse than the age-bin mean (1.086); with the donors in scope it reached 0.975 (PPCA + *k*NN 0.973): a small gain, not a failure. The population model was clearly worse than the norm when most of the scan was hidden (Fig. 6b; 1.454 above 50%), which the default `--max-missing-tracts` 0.5 excludes (the 0.5 default is thus needed for the PPCA-based settings and sufficient on BCP), and above the norm's error for the smallest masks (1.081 when less than 5% was hidden, 247 of 433 passes; 1.201 for single-tract donors). These cell-weighted points are driven by a minority of scans with large errors (heavy tails): per scan on the passes hiding less than 5%, PPCA + SIREN had 0.901 of the norm's error (CI 0.845–0.966) but was better on only 63% of scans, against 90% for the transformer (0.748) and 82% for the released default (0.760); over all passes in scope it had 0.861 (CI 0.820–0.902), better on 76% of scans. The cell-weighted scores of the intermediate classes (0.946, 0.792 and 0.973 for 5–10, 10–25 and 25–50% hidden; 86, 64 and 16 passes) should not be read as a U-shaped dependence on load: they rest on few passes and are dominated by the same tails. Second, the transformer, trained on six times as many scans as in BCP, was much stronger and robust to the load: alone it reached 0.578 overall (BCP, 400 epochs: 0.632), 0.740 for regions and 0.743 for the real masks in scope (0.829 above 50% hidden). The blends remained best for the synthetic patterns, including whole profiles (shipped table 0.612 vs 0.630 for the transformer alone; fixed blend 0.631). For regions and the real masks, the transformer alone matched or beat the blend with the shipped BCP table (real masks in scope 0.743 vs 0.768; per scan 0.993, CI 0.979–1.005), but a weight table refitted on HBCD (cross-fitted) matched or beat it: 0.545 overall against 0.550 for the shipped table on the same cells, and 0.737 on the real masks in scope (per scan 0.983 of the transformer alone's error, CI 0.977–0.989). Refitted on the target cohort, the table gives the stronger transformer more weight where it is better.

Pooling depends on age. Pooling HBCD into the norm and population model of BCP (batch-corrected) made PPCA + SIREN worse overall (1.029; Appendix Table A21), but not uniformly: it tended to help at 0–3 months (0.977), where BCP has 28 scans and HBCD 1,501, and hurt at 4–9 (1.057) and 10–60 months (1.039), where HBCD is sparse or differs in age composition within the bins. The fully pooled fixed blend (pooled norm, PPCA and transformer; 0.562) was equivalent to the released BCP-only default (0.998), better at 0–3 months (0.957), and 1.6% worse

Table 8: E3: HBCD alone (seeds 0–2, 20 scans per bin; transformer 200 epochs; shipped BCP weight table). Real mask: all donors (donors in the tool's scope: Table 7).

Setting	Overall (SD)	Gap	End	Whole	Region	Tensor	NODDI	Real mask
age-bin mean	1.000 (0.000)	1.000	1.000	1.000	1.000	1.000	1.000	1.000
PPCA alone	0.776 (0.020)	0.681	0.717	0.727	0.981	0.628	0.757	1.097
PPCA + kNN	0.703 (0.018)	0.557	0.602	0.682	0.970	0.628	0.757	1.083
PPCA + SIREN	0.644 (0.025)	0.409	0.499	0.690	0.978	0.628	0.757	1.086
Transformer alone	0.578 (0.004)	0.463	0.481	0.630	0.740	0.341	0.516	0.767
Fixed blend (SIREN base)	0.559 (0.006)	0.385	0.454	0.631	0.766	0.364	0.536	0.816
Fixed blend (kNN base)	0.609 (0.000)	0.498	0.538	0.635	0.763	0.364	0.536	0.818
Precision blend	0.578 (0.013)	0.401	0.444	0.629	0.841	0.375	0.563	0.901
Shipped table (SIREN base; released default)	0.550 (0.004)	0.389	0.444	0.612	0.756	0.341	0.519	0.799
Shipped table (kNN base)	0.592 (0.001)	0.476	0.511	0.622	0.757	0.341	0.519	0.805

than training only the transformer on both cohorts (1.016). At equal training length (200 epochs), the transformer trained on BCP + HBCD was 0.889 of the BCP transformer's error alone, and 0.957 within the fixed blend; the gain was largest at 0–3 months (0.865).

Pooling the population model, not the norm, causes the loss (E5). Pooling the two components separately (Table 9) located the loss. With the norm pooled and the PPCA fitted on BCP alone, the error was equivalent to BCP alone (0.999; 0–3 months 0.977, 4–9 1.006, 10–60 1.008). With the BCP norm and a PPCA fitted on both cohorts, it was 1.050 overall and 1.069 and 1.075 at 4–9 and 10–60 months (0–3 months 0.991). Pooling both reproduced the earlier result (1.029), and replacing the age bins by a continuous kernel norm did not remove the loss (1.032; 4–9 months 1.043, 10–60 months 1.049): the within-bin age composition is not the cause. The loss was larger for the real missingness masks: with the pooled PPCA, the error on the real-mask pattern was 1.156 times that with the BCP PPCA (CI 1.115–1.198; both pooled 1.164, CI 1.135–1.192). A population model learns how the deviations of a scan co-vary across tracts and measures; pooled with HBCD, whose scans are mostly under 10 months and come from other scanners, another preprocessing pipeline and a term-corrected age scale, it learns a covariance that fits the BCP scans worse, most of all the older ones the second cohort barely covers. A more specific hypothesis, which we did not test: HBCD is multi-site and multi-vendor, and with 84% of the pooled scans its between-site variance probably dominates the leading PPCA components; this would also explain why the PPCA fitted on HBCD alone gained little for regions (E3, 0.978). The norm itself, a mean and SD per cell and age bin, is shifted and scaled to BCP by the batch correction and does no harm.

The batch correction assumes one shift and scale per cell at all ages. After correction, the cohorts agreed within about 0.3 SD in most age windows, but at 1.5–3.5 months they differed by up to 2.12 SD (MD, RD; Appendix Table A12), where BCP has 20 scans and the cohorts' mean ages differ. The age-by-cohort interaction is thus not negligible in early infancy. The cohorts' age scales differ (BCP chronological, HBCD term-corrected), so the batch correction "at the same age" and the pooled age bins compare scans at slightly different developmental ages. This effect is largest in the youngest infants, where the measures change fastest, and is consistent with the residual difference at 1.5–3.5 months; without gestational ages we cannot show that it is the cause. The different acquisitions and preprocessing pipelines of the two cohorts are a further possible contributor.

4.8 Error by measure, age and tract

Measures. MD, RD and NDI were imputed best (released default 0.504, 0.504 and 0.505); FA, AD, ODI and FWF worst (0.610, 0.587, 0.645, 0.631; Appendix Table A18, Fig. 7). For a missing

Table 9: E5: pooling decomposition, PPCA + SIREN on BCP scans with BCP and HBCD pooled in different components, relative to BCP alone on the same scans (overall score; paired ratio with 95% subject-bootstrap CI, all and per age bin).

PPCA + SIREN with	Overall	All (paired)	0–3 mo	4–9 mo	10–60 mo
Pooled norm, BCP PPCA	0.610	0.999 [0.992, 1.006]	0.977 [0.955, 0.999]	1.006 [1.000, 1.013]	1.008 [1.002, 1.013]
BCP norm, pooled PPCA	0.644	1.050 [1.035, 1.066]	0.991 [0.962, 1.018]	1.069 [1.050, 1.090]	1.075 [1.054, 1.096]
Both pooled, age bins	0.626	1.029 [1.014, 1.044]	0.977 [0.939, 1.011]	1.057 [1.035, 1.082]	1.039 [1.025, 1.052]
Both pooled, kernel age norm	0.630	1.032 [1.019, 1.045]	0.991 [0.957, 1.027]	1.043 [1.025, 1.065]	1.049 [1.037, 1.063]
Both pooled, age bins (v15)	0.626	1.029 [1.014, 1.044]	0.977 [0.939, 1.011]	1.057 [1.035, 1.082]	1.039 [1.025, 1.052]

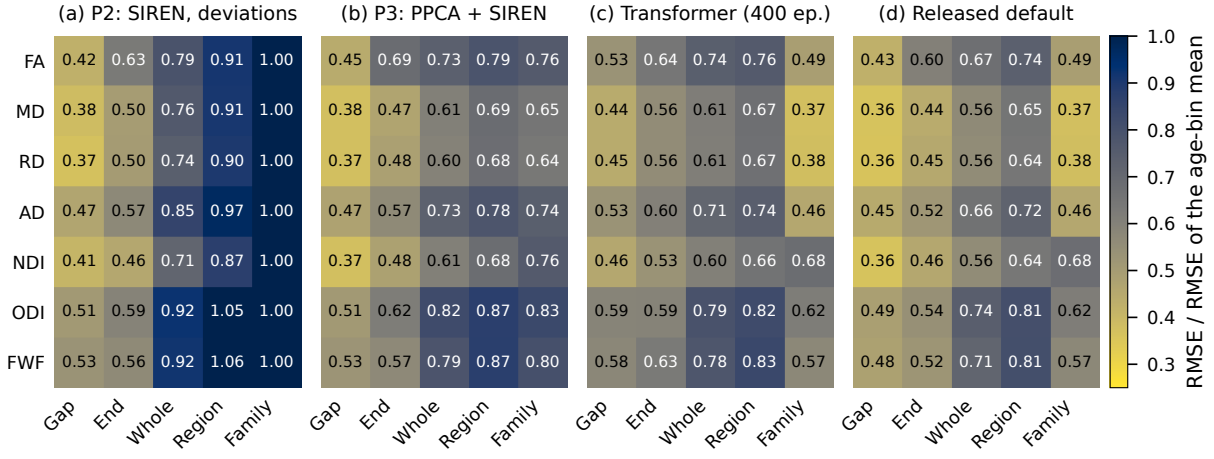


Figure 7: Error relative to the age-bin mean per measure and pattern (mean over age bins and 3 seeds) for four settings. "Family": the measure's family hidden (1.00 for the norm-referenced SIREN, which cannot impute it).

family the tensor measures were recovered well from NODDI (MD 0.374, FA 0.490), the NODDI measures less well from the tensor (NDI 0.678, ODI 0.615, FWF 0.574).

Age. Relative to the age-bin mean the error varied little with age (Fig. 8a, b; PPCA + SIREN 0.610, 0.610 and 0.622 in the three bins); the ranking of the methods was the same in every age group, and the youngest and oldest groups have wide intervals (few scans). In absolute terms the age-bin mean's diffusivity errors were largest at 0–3 months, and the released default halved them (Appendix Table A19, Fig. 8c).

Tracts. The whole-profile error of the released default ranged from 0.425 (SLF_II_R) to 0.832 (AC_olfactory) across tracts (Fig. 9); 2 tracts were above 0.8 and none above 1. Tracts far from other tracts were imputed worse (Spearman 0.40, $p = 6.2\times10^{-4}$); tract length did not matter.

4.9 Uncertainty

The SD that `impute-profiles` writes for a blended value combines the PPCA and transformer SDs with the blend weights. For the released default (Appendix Tables A5 and A6), the root-mean-square of error/SD was 0.86 for whole profiles and 0.96 for regions (97% and 96% of errors within 1.96 SD), 1.10 for a missing family (slightly too small SDs), and 0.55 within 0.5 mm of an observed value (SDs about twice too large: the PPCA SD is an upper bound for locally imputed values); beyond 4 mm it was 0.98. The HBCD-reference blend was better calibrated for a missing family (0.93). Users can treat the SDs as conservative near observed values and as about right elsewhere; a per-band rescaling is in Appendix Table A6.

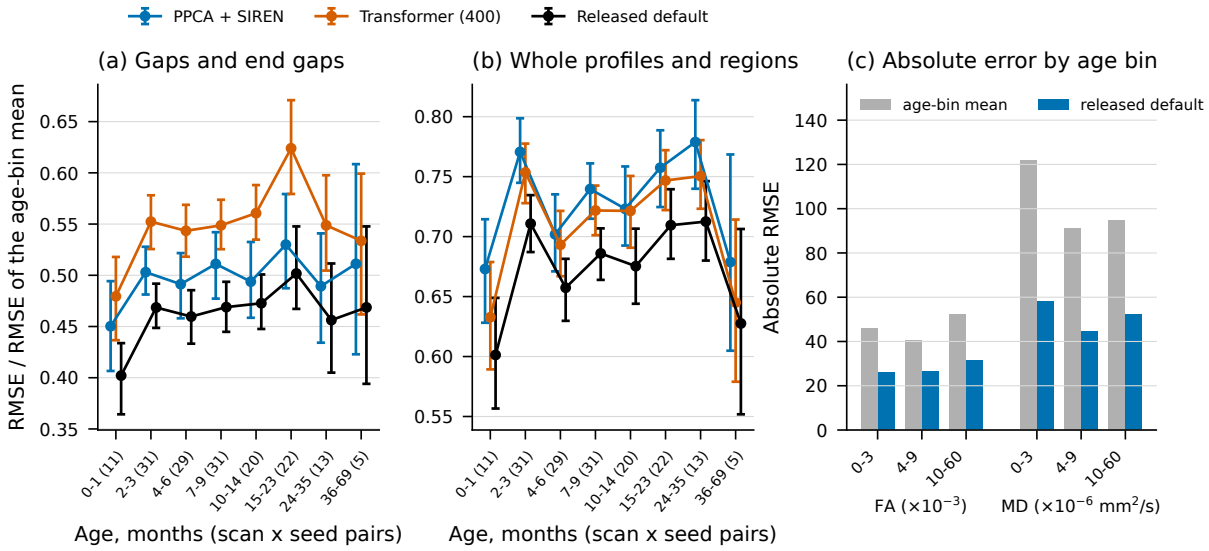


Figure 8: Error by age. (a, b) Error relative to the age-bin mean per age group (cell dumps, 3 seeds; error bars: 95% bootstrap CI over scan × seed pairs, their number in parentheses) for PPCA + SIREN, the transformer alone and the released default. (c) Absolute RMSE of FA and MD of the age-bin mean and the released default per age bin.

4.10 Downstream analyses

Table 10 and Fig. A3 summarise 50 replicates per scenario. With the real prevalence of missing values, complete-case analysis kept 114 of the 220 scans: its trajectory CIs were 1.328 times as wide as with complete data, its curves deviated more from the complete-data curves (0.047 SD; with random masks this is the sampling deviation of a fit on half the scans, not a bias) and, for a planted effect of 0.1 SD, its power fell from 36% to 26%. Every imputer kept all scans and reproduced the complete-data analysis closely: trajectory deviation 0.003 SD (age-bin mean 0.010), CI width 0.997, coverage 100%, power 98% for $\delta = 0.3$ and 35% for $\delta = 0.1$, type I error 5.6% (cluster-robust SEs with 133 subjects are slightly liberal for every imputer, complete-case analysis included). Imputation attenuated a planted group effect only slightly (estimate 0.980 of the complete-data estimate for the blend, 0.980 for PPCA, 0.982 for the age-bin mean; difference -0.006 SD), because only a few percent of each tract's values were missing (2.1%). At $\delta = 0.1$ the ratio of estimates is a ratio of small, noisy numbers (complete case 1.177); the differences in SD units (Table 10) are the more reliable summary. When every scan carried a real mask, the age-bin mean attenuated the effect more (0.938) than PPCA (0.993) or the blend (0.991), and complete-case analysis was impossible. Multiple imputation from the PPCA posterior gave the same estimates and SEs as point imputation (SE ratio 0.999): at this level of missingness, the between-imputation variance is negligible. Tract-wise available-case analysis was nearly as good as imputation for single-tract analyses but cannot serve analyses of all tracts jointly.

These scenarios assign the masks independently of group and values. In two further scenarios with differential missingness, every scan of the group carrying the planted effect received a real mask and no scan of the other group (about twice the cohort-wide share of imputed values in that group, 2.2% of each tract's values overall). Complete-case analysis then removed the whole effect group and could not estimate the group effect. Every imputer attenuated the effect by about the share of imputed values in the effect group: estimate 0.956 of the complete-data estimate for the blend, 0.956 for PPCA and 0.959 for the age-bin mean at $\delta = 0.3$ (difference -0.012 SD), with unchanged power (98%) and type I error (5.7%). Because the effect was planted in whole tracts, the imputed values carry no information about it, and no imputer recovered it: imputation toward

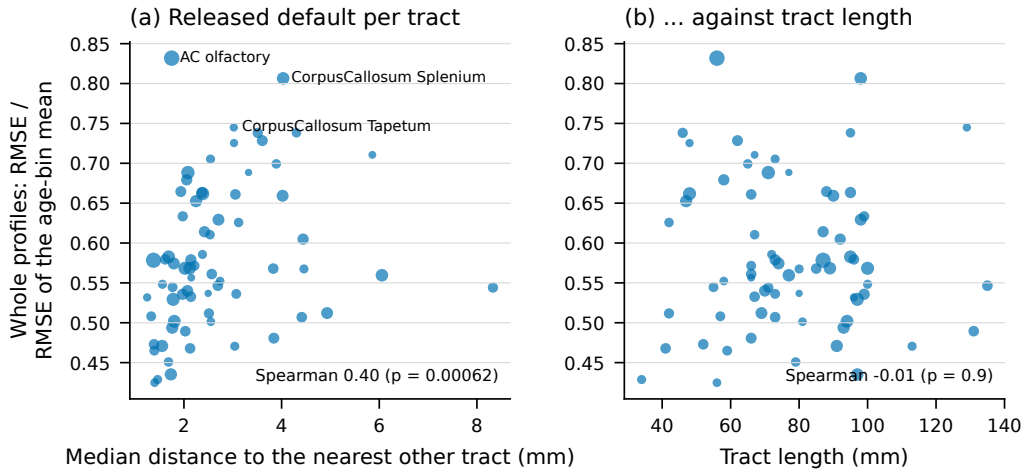


Figure 9: Whole-profile error of the released default per tract (seeds 0–2) against (a) the median distance to the nearest other tract and (b) the tract length; marker area $\propto$ the tract's real rejection rate.

the norm shrinks a group difference in proportion to the imputed share of the affected group. At the missingness of these cohorts the shrinkage is a few percent; it grows with that share.

4.11 Statistical summary

Appendix Table A20 and Appendix Fig. A5 collect the selected four-pattern contrasts. Under the subject-level bootstrap, the CIs were nearly identical to the per-pair CIs. Differences of 1–2% (components, blend weights, epochs) were detectable because all settings saw the same hidden values; their practical relevance is small, and they were selected on the same draws.

5 Discussion

5.1 Principal findings

1. *A scan's own neighbouring tracts do not predict a missing tract.* Same-scan interpolation is useful only along a profile.
2. *The normative reference is the main source of information.* Imputing deviations from an age-appropriate norm made every pattern imputable; a linear population model of how deviations co-vary across tracts and measures improved whole profiles and regions further and made a missing measure family imputable.
3. *The best local refinement depends on the pattern.* For whole missing tracts, the dominant real pattern, kNN on the population residuals is better than SIREN; SIREN wins for gaps.
4. *Amortized networks add most where local information is scarce*, beyond a few millimetres and for a missing family, when they correct the population model and are blended with it.
5. *A second cohort helps as training data for the transformer, not inside the population model.* Training the transformer on both cohorts reduced its error (0.889 at equal epochs). Pooling the norm was harmless, but pooling the PPCA multiplied the error by 1.050 on the synthetic patterns and by 1.156 on the real masks, with either age norm. The different age scales (BCP chronological, HBCD term-corrected), acquisitions and preprocessing pipelines of the cohorts are plausible reasons why their covariance structures differ.

Table 10: E4: downstream analyses on 220 complete BCP scans with real masks transplanted (50 replicates per scenario). Trajectory: FA and MD tract means ∼ spline(log age) on the tracts without a planted effect, against the complete-data fit; deviation: mean absolute difference from the complete-data curve (SD of the tract mean). Group effect: planted in 5 tracts; Est. − full: mean difference from the complete-data estimate (between-scan SD); type I: rejections in the tracts without a planted effect. "Group 1 masked": differential missingness, masks only in the group with the planted effect.

		Trajectories (no effect planted)				Planted group effect			
Scenario	Imputer	Deviation (SD)	CI width	Coverage	Type I	Est. − full (SD)	Est. / full	SE / full	Power
$\delta = 0.3$, prevalence	complete case	0.047	1.33	99%	5.6%	+0.001	1.008	1.26	94%
	available case	0.006	1.01	100%	5.3%	+0.000	1.001	1.01	98%
	age-bin mean	0.010	1.00	100%	5.1%	-0.006	0.982	0.99	98%
	PPCA	0.003	1.00	100%	5.6%	-0.006	0.980	1.00	98%
	PPCA + kNN	0.003	1.00	100%	5.6%	-0.006	0.981	1.00	98%
	blend (table)	0.003	1.00	100%	5.6%	-0.006	0.980	1.00	98%
	multiple imputation (PPCA)	0.003	1.00	100%	5.5%	-0.006	0.980	1.00	98%
$\delta = 0.1$, prevalence	complete case	0.047	1.33	99%	5.6%	+0.001	1.177	1.26	26%
	available case	0.006	1.01	100%	5.3%	+0.000	1.045	1.01	37%
	age-bin mean	0.010	1.00	100%	5.1%	-0.002	1.007	0.99	36%
	PPCA	0.003	1.00	100%	5.6%	-0.002	0.982	1.00	34%
	PPCA + kNN	0.003	1.00	100%	5.5%	-0.002	0.975	1.00	35%
	blend (table)	0.003	1.00	100%	5.5%	-0.002	0.982	1.00	35%
	multiple imputation (PPCA)	0.003	1.00	100%	5.6%	-0.002	0.985	1.00	34%
$\delta = 0.1$, all masked	complete case				every scan excluded				
	available case	0.010	1.02	100%	5.4%	-0.000	1.043	1.01	35%
	age-bin mean	0.020	1.00	100%	5.4%	-0.005	0.938	0.99	35%
	PPCA	0.005	0.99	100%	5.5%	-0.004	0.993	0.99	34%
	PPCA + kNN	0.005	1.00	100%	5.5%	-0.004	0.994	0.99	33%
	blend (table)	0.005	0.99	100%	5.5%	-0.004	0.991	0.99	34%
	multiple imputation (PPCA)	0.005	0.99	100%	5.4%	-0.004	0.991	0.99	34%
$\delta = 0.3$, group 1 masked	complete case	0.058	1.41	98%	–	–	–	–	–
	available case	0.006	1.01	100%	5.5%	-0.000	1.000	1.01	98%
	age-bin mean	0.011	1.00	100%	5.6%	-0.012	0.959	1.00	98%
	PPCA	0.003	1.00	100%	5.5%	-0.013	0.956	1.00	98%
	PPCA + kNN	0.003	1.00	100%	5.6%	-0.012	0.957	1.00	98%
	blend (table)	0.003	1.00	100%	5.7%	-0.012	0.956	1.00	98%
	multiple imputation (PPCA)	0.003	1.00	100%	5.5%	-0.013	0.956	1.00	98%
$\delta = 0.1$, group 1 masked	complete case	0.058	1.41	98%	–	–	–	–	–
	available case	0.006	1.01	100%	5.5%	-0.000	1.025	1.01	36%
	age-bin mean	0.011	1.00	100%	5.6%	-0.003	0.997	1.00	35%
	PPCA	0.003	1.00	100%	5.5%	-0.004	0.975	1.00	34%
	PPCA + kNN	0.003	1.00	100%	5.6%	-0.004	0.987	1.00	33%
	blend (table)	0.003	1.00	100%	5.6%	-0.004	0.980	1.00	34%
	multiple imputation (PPCA)	0.003	1.00	100%	5.5%	-0.004	0.973	1.00	34%

6. *The gains hold on untouched subjects and real masks.* On subjects never used for any design decision, the released default had 0.925 of PPCA + SIREN's error (development: 0.923), with the same ranking for whole profiles and for the real missingness masks. On a second cohort (HBCD alone) the blend's gain was larger (0.872). With a large training cohort the transformer alone matched or beat the blend with the shipped BCP weights for regions and real masks, but not for whole profiles, and a weight table refitted on the target cohort matched or beat it everywhere. The population model gained little on HBCD's real masks (0.975 within the tool's scope, cell-weighted; per scan 0.861, better than the norm in only 76% of scans against 98% for the transformer), was above the norm's cell-weighted error for the smallest masks (1.081) through a minority of poorly imputed scans, and was clearly worse than the norm when most of a scan was hidden; for a large multi-site cohort the transformer is the safer choice.

7. *At realistic missingness, imputation preserves typical analyses and complete-case analysis does not.* With masks independent of group and values, imputation kept all scans with negligible bias and attenuation; complete-case analysis lost about half of the scans, and its power for a 0.1-SD group effect fell from 36% to 26%. With all masks in the group carrying the effect, every imputer attenuated the group effect by about the imputed share of that group (0.956 of the complete-data estimate), without inflating type I error. Multiple imputation was not needed at this level of missingness.

5.2 Interpretation

The PPCA posterior is the exact conditional expectation under a Gaussian model with one loading per cell; with 80 components it captures the main covariation of 37,716 cells from a few hundred scans. Functa and the partial VAE replace this exact inference by a latent fitted with a few gradient steps or an encoder, and with hundreds of scans they did not learn a better prior. The networks that read each location's observed value directly and only learn a correction to the PPCA prediction were better, most where PPCA has least local information. We offer some hypotheses that our data do not test: that neighbouring tracts differ systematically by their orientation and organization (which would explain the failure of same-scan methods), that the transformer's global attention explains its advantage over the U-Net (the training budget does not, O5), and that context inputs did not help because the observed values already encode age and acquisition.

5.3 Limitations

- *One normative cohort for development.* Choices were made on seeds 0–2; E1 tests them on untouched subjects of 4–60 months only (the 0–3 month bin was exhausted), and E3 on a second cohort (with the transformer trained for 200 instead of 400 epochs).
- *Missingness mechanism.* The synthetic patterns hide values at random within each pattern; the real-mask pass reproduces the real masks but not a possible dependence between rejection and the true values (missing not at random).
- *QC truncates the hidden values.* The profile QC removed tracts that deviated by more than 3.5 SD from a prior built on the full BCP cohort, so only values that passed this outlier filter could be hidden. The errors are therefore likely optimistic for atypical scans, such as those of clinical cohorts or with residual artefacts.
- *Reference of the score.* Ratios to the age-bin mean depend on how good that mean is; the kernel-norm reference and absolute RMSEs (Table 4, Appendix Table A4) give the same ranking. BCP ages are nominal whole months.
- *Age scales of the cohorts.* BCP ages are chronological, HBCD ages term-corrected. For infants born before term the scales differ by $40 - \text{GA}$ weeks, so pooled norms, the batch correction and the 0–3 month results compare slightly different developmental ages. This plausibly contributes to the residual cohort difference at 1.5–3.5 months and to the age dependence of pooling, but gestational age is not in our data and we could not test it. Correcting both cohorts to a common age scale (e.g. postmenstrual age) before pooling is left for future work. The cohorts were also acquired differently and preprocessed with different pipelines (`dmriprep`, QSIPrep; the model fitting and later steps were the same), which may add cohort differences beyond a shift and scale per cell.
- *NODDI in infants.* NODDI was fitted with AMICO's default (adult) intrinsic diffusivities, which differ from those of the infant brain (Cieslak et al., 2025); NDI, ODI and FWF are therefore model-dependent indices whose absolute values should be interpreted with care, although this does not affect the comparison of imputation methods.
- *Dependence and training variance.* The subject bootstrap handles repeated seeds and sessions but not the variation from network training; some checks used one seed or one split.
- *Measures and atlas.* Seven measures of one atlas with dense tract axes; sparser tract sets give fewer close neighbours.

5.4 Practical recommendations

1. *Without a normative reference*, fill only gaps along a profile (arc-linear); do not impute whole tracts from the scan's own neighbours.
2. *With a normative cohort and whole missing tracts (the common case)*, fit a population model (PPCA, about 80 components) of the deviations from an age-appropriate norm and refine it locally with kNN on its residuals within 4 mm; this needs no GPU. The error falls steadily with the size of the normative cohort, without a threshold: relative to about 395 scans, the whole-profile error was 1.187 times as large with about 99, 1.073 with about 198 and 1.022 with about 295 (norm and PPCA refitted on each subset; Section 4.3). In a large multi-site cohort (HBCD) the population model gained little for spatially clustered missingness (regions 0.970, real masks 0.973 for PPCA + kNN), so a GPU-trained transformer is worth its cost there. Keep the three broad age bins: finer bins did not help under the population model.
3. *If short gaps along profiles matter*, use SIREN on the residuals instead (a combination of kNN for whole tracts and SIREN within 2 mm for gaps is plausible but was not evaluated).
4. *With a GPU*, blend with a transformer trained to correct the population model, with weights growing with the distance to the nearest observed value (the default of DTIPlayground 0.9.8); this is most valuable when whole measure families are missing. With a large normative cohort (thousands of scans, as HBCD), the transformer alone matched or beat the blend with the shipped BCP weights for regions and real masks, though not for whole profiles; a weight table refitted on the target cohort (`--ensemble-weightsfit`) matched or beat the transformer alone for every pattern, so refit the weights, or check the blend against the transformer alone in a validation of the target cohort.
5. *With a second cohort on the same tracts*, use it to train the transformer, and keep the population model of the target cohort; the norm can be pooled after a batch correction (it neither helped nor hurt overall, 0.999, and helped slightly in the thin 0–3 month bin, 0.977 [0.955, 0.999]; E5, Table 9), preferably after bringing both cohorts to a common age scale (chronological vs term-corrected or postmenstrual age).
6. *Report and check*: report the share of imputed values, per group, keep the masks and SDs, and check that conclusions hold without imputed values. At the levels of missingness seen in BCP and with missingness unrelated to group, point imputation preserved trajectory and group analyses and was much better than complete-case analysis. When one group has more rejected tracts (a clinical or high-motion group), imputation toward the norm shrinks the group difference by about the imputed share of that group: compare group results with tract-wise available-case analysis, and do not rely on imputed values for group differences between normative and clinical scans, whose deviations the norm of a typically developing cohort does not represent. With much more missingness, prefer the population model or the blend over the age-bin mean (less attenuation when every scan is masked) and consider multiple imputation from the PPCA posterior.

6 Conclusion

In a developmental cohort with dense atlas tracts, missing diffusion profile values are best imputed as deviations from an age-appropriate norm, using a joint population model of all tracts and measures, refined locally near observed values and, with a GPU, blended with a transformer trained to correct it. Several more complex alternatives did not help. The gains reproduced on untouched subjects and real missingness masks, and at the missingness of a typical infant

cohort imputation preserved downstream analyses that complete-case analysis would weaken; with missingness concentrated in one group it shrinks that group's difference by about its imputed share. On a second, larger infant cohort, the ranking transferred, the amortized transformer gained most, and blend weights refitted on that cohort did best. All methods, the validation protocol and the defaults are available in DTIPlayground.

Data and code availability

DTIPlayground (including `dmrifiberprofileimpute-profiles` and its `--validate` protocol) is open source at https://github.com/NIRALUser/DTIPlayground, on PyPI (`dtiplayground`; this paper: version 0.9.9, GitHub tag v0.9.9) and on Docker Hub (`niraluser/dtiplayground:0.9.9`) [TODO: Zenodo DOI of the release]. The scripts that compute every number, table and figure, together with the manuscript sources, are in the same repository under `papers/ProfileImputation` (`scripts/run_all.sh`). A Zenodo dataset (Styner et al., 2026) (https://doi.org/10.5281/zenodo.24110365) contains aggregate data only: the normative age-bin statistics (mean, SD, N and quartiles) of BCP and HBCD, the aggregated validation results of all runs (by pattern, measure, age bin, tract and distance), the aggregate tables behind the paper's tables and figures, and the processing code (model fitting, registration, registration QC, profile extraction and profile QC, and the `dmriprep` protocol). Participant-level profiles and per-scan results of neither cohort are redistributed; the paired per-scan statistics can be recomputed only with access to the cohorts. Whether the trained models (population model, transformer) can be shared is being clarified with the data providers [TODO: author: model sharing]. BCP data are available through the NIMH Data Archive (NDA) [TODO: author: NDA collection ID, if cited]; the profiles used here were processed by the BCP imaging core from the study's imaging data. HBCD data (release v2.0) are available through the NIH Brain Development Cohorts (NBDC) Data Sharing Platform under its data use agreement [TODO: author: DUA or study number, if required].

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Author contributions

[TODO: CRediT statement.]

Declaration of competing interests

[TODO: Competing interests.]

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A Supplementary material

A.1 Acquisition, preprocessing and processing

BCP. The BCP is a joint study of the University of North Carolina at Chapel Hill and the University of Minnesota that enrols typically developing children between birth and 5 years of age in a hybrid accelerated longitudinal and cross-sectional design (planned: 500 children, 285 of them with 4–6 longitudinal visits) (Howell et al., 2019). Children were eligible if born at 37–42 weeks of gestation with a birth weight appropriate for gestational age and without major pregnancy and delivery complications; exclusion criteria included a first-degree relative with autism, intellectual disability, schizophrenia or bipolar disorder, significant medical or genetic conditions affecting growth, development or cognition, and contraindications to MRI (Howell et al., 2019). Children were scanned without sedation, the younger ones during natural sleep, on 3T Siemens Prisma scanners with 32-channel head coils at both sites. The diffusion protocol of Howell et al. (2019) is a 6-shell scheme (b = 500, 1000, 1500, 2000, 2500 and 3000 s/mm^2 with 9, 12, 17, 24, 34 and 48 directions, 6 b = 0 images), acquired at 1.5 mm isotropic resolution with multiband factor 5 in both the AP and PA phase-encoding directions (6:58 each); in children who stayed asleep, a 2-shell scheme with 79 directions (b = 1500 and 3000 s/mm^2, multiband factor 4, 1.5 mm isotropic, AP and PA, 4:35 each) was added, and was prioritized for longitudinal participants of an earlier study to keep their data compatible. Parents provided permission and informed consent, and the procedures were approved by the Institutional Review Boards of both universities (Howell et al., 2019). The labels of the scans give the sequence: `dir79select` (311 scans) the 79-direction 2-shell sequence, `shells06select` (108) and `shells06` (4) the 6-shell sequence (by age bin in Appendix Table A13). Some sessions have only one sequence, others both; a session with both contributes two scans, each processed and profiled separately (39 of 384 sessions). The BCP scans were preprocessed with `dmriprep` of DTIPlayground (version 0.5.7; NIRAL, 2026) with a protocol of five modules: a slice-wise intensity check and an interlace (inter-slice correlation and motion) check that exclude corrupted gradient volumes, as in DTIPrep (Oguz et al., 2014); susceptibility-distortion correction with FSL's topup from the reverse phase-encoded acquisitions (Andersson et al., 2003); eddy-current and head-motion correction with FSL's eddy, with susceptibility-by-movement estimation and replacement of outlier slices (Andersson and Sotiropoulos, 2016); and a weighted least-squares tensor fit; a QC report lists the excluded gradients and the motion of every scan. The BCP data were processed by the BCP imaging core (UNC) from the study's own imaging data.

HBCD. HBCD scans were acquired at multiple sites on Siemens, GE and Philips scanners with two complete multi-shell acquisitions of opposite phase-encoding direction (AP and PA), each with b = 0, 500, 1000, 2000 and 3000 s/mm^2 (10, 6, 12, 18 and 30 volumes), 87 contiguous 1.7-mm axial slices with a 140 × 140 matrix and a 238-mm field of view, multiband factor 3, TR 4800 ms and TE 88 ms (Dean et al., 2024; Cieslak et al., 2025). They were preprocessed with QSIPrep as described by Cieslak et al. (2025): partial-Fourier Gibbs unringing, MP-PCA denoising, susceptibility-distortion, eddy-current and head-motion correction with FSL's topup and eddy followed by a second distortion correction with DRBUDDI (TORTOISE), brain extraction with SynthStrip on the T2-weighted image, and resampling to an AC-PC space aligned with the T2-weighted image. The derived maps of QSIRecon were not used; the diffusion models were fitted to the QSIPrep outputs as described below.

Processing. Both cohorts then went through the same steps. Model fitting used DTIPlayground 0.5.10; registration, registration QC, profile extraction and profile QC used DTIPlay-

ground 0.9.0. (i) *Models.* Within the brain mask (for HBCD that of QSIPrep), the diffusion tensor was fitted by weighted least squares with DIPY (Garyfallidis et al., 2014) on the volumes with $b \leq 1600\,\mathrm{s/mm^2}$: for HBCD the shells of 500 and $1000\,\mathrm{s/mm^2}$, for the BCP 6-shell sequence those of 500 to $1500\,\mathrm{s/mm^2}$ and for the 79-direction sequence the shell of $1500\,\mathrm{s/mm^2}$; NODDI (Zhang et al., 2012) was fitted with AMICO (Daducci et al., 2015) on all shells, with AMICO's default model settings (no infant-specific intrinsic diffusivities were set; 2000 kernel orientations); and the free-water corrected tensor (Pasternak et al., 2009; Hoy et al., 2014) with DIPY (non-linear least squares, all shells), whose measures were not used here. (ii) *Registration.* Each tensor image was registered with the DTI_Register module of `dmriprep` (DTI-Reg; Verde et al., 2014): an affine registration of the FA images (BRAINSFit) refined by a greedy diffeomorphic ANTs registration with a cross-correlation metric (Avants et al., 2008), the tensors resampled log-Euclidean with reorientation (Arsigny et al., 2006), to the age-appropriate mean tensor of a normative model of the pediatric DTI_IBISEP_Feb26 atlas (one mean tensor per age bin, on the atlas grid), after an automatic check and correction of the tensor frame. The atlas was built from IBIS data starting from the UNC pediatric atlas (Short et al., 2022); its normative model (mean tensors and their spread per age bin), used for the registration and the registration QC, and the prior age-bin profile statistics used in the profile QC were built with DTIPlayground from the full BCP cohort. These full-cohort statistics enter only the processing and QC stage, which defined the real missingness, and the norm of the released imputation default; the validation does not use them: in every validation run the age-bin norm, the population models and the networks were refitted from the cohort itself without the validated subjects (leave-subject-out folds; no validation setting used the precomputed statistics). (iii) *Registration QC.* `dmrifiberprofileqc-registration` compared each registered scan with the atlas and the age-binned normative model (FA and the diffusivities: mean absolute error, structural similarity, correlation, principal-direction angular error, contiguity of the disagreement, and a check of the ventricles) and flagged a session as a registration failure by a robust Mahalanobis distance over these features. (iv) *Profiles.* Profiles were extracted with the EXTRACT_Profile module of `dmrifiberprofile` in each scan's native space: the arc-length-parametrized fibers of each atlas tract were mapped into the scan through its deformation field, so that the tensors did not need to be resampled into atlas space; FA, MD, RD and AD were computed from the tensors interpolated log-Euclidean along the fibers, NDI, ODI and FWF sampled from their maps. The profile was sampled every 1 mm of the stored atlas arc length, from an origin at the tract's median plane, so that all scans share the atlas's sampling grid, and points within a Gaussian kernel of 3 mm SD around each sample were averaged. The profile QC that follows is described in Section 3.1.

A.2 Supplementary tables and figures

Table A1: Development history: the phases in which the methods were added to DTIPlayground, and where they appear here.

Phase (release)	Methods	Family here
1	arc-linear, linear, kNN, SIREN on the scan's own values	A
2 (0.8.20)	the same on deviations from the age-bin norm	B
3 (0.8.21)	PPCA, PPCA + kNN / SIREN (default up to 0.9.5)	C
3b (0.8.22)	continuous age norms, pooling with batch correction	C, second cohort
4 (0.8.23)	graph U-Net after CONFETTI	E
5	functa (not adopted)	G
6 (0.9.8)	transformer, partial VAE, age-dependent PPCA, acquisition correction, blends	E, F, G
revision	generic cohort imputers, untouched test, real masks, HBCD, downstream	D, Sections 4.5–4.10

Table A2: Generic cohort imputers (E8) on the hidden values of seeds 0–2 (same training scans per fold as PPCA), with the paired whole-profile ratio against PPCA alone (95% subject-bootstrap CI).

Setting	Overall (SD)	Gap	End	Whole	Region	Tensor	NODDI	Whole: paired vs PPCA
offset (measure)	0.903 (0.011)	0.870	0.939	0.887	0.917	1.002	1.000	1.218 [1.194, 1.244]
offset (5 tracts)	0.898 (0.007)	0.853	0.927	0.876	0.936	1.002	1.000	1.202 [1.181, 1.226]
kNN across scans, k=5	0.856 (0.011)	0.822	0.898	0.838	0.867	0.819	0.936	1.167 [1.150, 1.185]
kNN across scans, k=10	0.838 (0.009)	0.811	0.877	0.821	0.843	0.803	0.914	1.140 [1.125, 1.157]
kNN across scans, k=20	0.837 (0.011)	0.811	0.872	0.821	0.842	0.802	0.910	1.134 [1.118, 1.151]
kNN across scans, k=50	0.854 (0.010)	0.829	0.886	0.839	0.862	0.830	0.918	1.152 [1.134, 1.171]
iterative SVD, rank 10	0.789 (0.014)	0.758	0.839	0.773	0.786	0.768	0.857	1.083 [1.070, 1.097]
iterative SVD, rank 40	0.752 (0.012)	0.716	0.798	0.732	0.764	0.721	0.810	1.023 [1.016, 1.031]
iterative SVD, rank 80	0.736 (0.008)	0.688	0.781	0.715	0.758	0.698	0.800	1.001 [0.999, 1.003]
iterative SVD, rank 160	0.727 (0.007)	0.669	0.760	0.709	0.769	0.674	0.786	0.990 [0.979, 1.000]
MICE (tract means)	0.843 (0.016)	0.813	0.906	0.820	0.831	0.868	0.910	1.146 [1.128, 1.165]
PPCA alone	0.733 (0.008)	0.685	0.777	0.713	0.759	0.696	0.798	–
PPCA + kNN	0.664 (0.008)	0.568	0.659	0.675	0.752	0.696	0.798	–
PPCA + SIREN	0.614 (0.007)	0.440	0.552	0.697	0.766	0.696	0.798	–

Table A3: Distance limit of the local methods under the current protocol: the local method fills cells up to the limit, PPCA beyond (normalized squared errors of the same cells pooled over age bins, relative to the age-bin mean; mean over measures).

Local method	Limit	Gap	End	Whole	Region	Overall
SIREN	PPCA only	0.652	0.754	0.688	0.728	0.706
	2 mm	0.486	0.605	0.659	0.722	0.618
	4 mm	0.416	0.539	0.665	0.731	0.588
kNN	PPCA only	0.652	0.754	0.688	0.728	0.706
	2 mm	0.558	0.655	0.649	0.718	0.645
	4 mm	0.532	0.634	0.637	0.717	0.630

Table A4: Absolute RMSE of the main settings (four patterns; mean over patterns, age bins and seeds); FA, NDI, ODI, FWF $\times 10^3$, MD, RD, AD $\times 10^6\,\mathrm{mm}^2/\mathrm{s}$.

Setting	FA	MD	RD	AD	NDI	ODI	FWF
Age-bin mean	46.3	102.6	101.6	125.8	65.5	51.8	56.2
Kernel age norm (alone)	40.7	90.4	86.4	120.1	57.3	50.4	55.3
Phase 2: SIREN on deviations	31.5	64.9	63.0	89.7	40.1	38.7	43.5
PPCA alone	36.1	67.2	65.7	95.1	43.2	41.9	45.7
PPCA + kNN	32.2	60.5	59.4	87.0	37.9	38.5	41.9
PPCA + SIREN	30.5	54.4	53.7	80.4	35.1	35.9	39.0
Graph U-Net alone	32.2	61.6	60.3	87.6	39.7	39.0	42.5
Transformer alone (400)	30.7	58.4	57.3	81.4	36.9	35.6	39.9
Blend, fixed (400)	28.3	51.7	50.9	75.1	33.3	33.3	36.4
Released default (table)	28.0	51.5	50.7	73.9	33.1	32.8	35.7
HBCD-trained transformer, table	26.5	48.8	47.8	70.3	31.6	30.8	33.5

Table A5: Calibration of the predicted SDs: root-mean-square of error/SD (1: right; below 1: SDs too large) and share of errors within 1.96 SD (95%: right), mean over measures, seeds 0–2 (cell dumps).

Setting	Gap	End	Whole	Region	Family
PPCA + SIREN	0.59 / 99%	0.78 / 97%	0.93 / 96%	1.06 / 95%	1.06 / 95%
Transformer (400)	0.97 / 96%	1.02 / 96%	1.08 / 94%	1.08 / 94%	1.20 / 94%
Released default (table)	0.58 / 99%	0.72 / 98%	0.86 / 97%	0.96 / 96%	1.10 / 95%
Table, shipped weights	0.58 / 99%	0.72 / 98%	0.86 / 97%	0.96 / 96%	1.10 / 95%
HBCD-reference table blend	0.58 / 99%	0.68 / 98%	0.82 / 98%	0.89 / 97%	0.93 / 97%

Table A6: Root-mean-square of error/SD per weight-table distance band (all patterns); dividing the written SDs by these factors calibrates them.

Setting	0-0.5	0.5-1	1-2	2-3	3-4	4-6	6-10	10-20	>20	none
PPCA + SIREN	0.59	0.61	0.77	0.94	1.06	1.04	1.10	1.06	1.21	1.06
Transformer (400)	1.01	0.99	1.02	1.04	1.06	1.10	1.12	1.09	1.14	1.20
Released default (table)	0.55	0.56	0.71	0.83	0.92	0.98	1.02	0.99	1.08	1.10
HBCD-reference table blend	0.53	0.56	0.70	0.80	0.87	0.90	0.93	0.91	0.96	0.93

Table A7: Blend learners on the cell dumps, cross-fitted per seed (mean over seeds 0–2).

Transformer	Blend	Overall	Gap	End	Whole	Region	Tensor	NODDI
BCP transformer (400 ep.)	PPCA + SIREN	0.614	0.440	0.552	0.697	0.766	0.696	0.798
	transformer alone	0.632	0.510	0.588	0.694	0.737	0.425	0.623
	fixed 0.25/0.75	0.576	0.414	0.509	0.654	0.726	0.443	0.630
	weights per measure x distance (finer bands)	0.569	0.418	0.506	0.639	0.715	0.425	0.623
	calibrated precision weights	0.572	0.423	0.503	0.643	0.721	0.471	0.667
	gradient-boosted weight per cell	0.569	0.418	0.504	0.638	0.715	0.425	0.624
HBCD-reference transformer (200 ep.)	PPCA + SIREN	0.614	0.440	0.552	0.697	0.766	0.696	0.798
	transformer alone	0.580	0.456	0.532	0.633	0.698	0.432	0.662
	fixed 0.25/0.75	0.557	0.400	0.495	0.634	0.697	0.439	0.642
	weights per measure x distance (finer bands)	0.541	0.398	0.481	0.604	0.680	0.429	0.646
	calibrated precision weights	0.542	0.398	0.473	0.611	0.687	0.469	0.650
	gradient-boosted weight per cell	0.538	0.398	0.474	0.602	0.678	0.431	0.637

Table A8: Graph U-Net development (seed 0, 54 scans, 11 measures): from CONFETTI's configuration to the hybrid.

Setting (seed 0, 54 scans, 11 measures)	Overall	Gap	End	Whole	Region	Family
CONFETTI configuration: fiber graph, 64 features	0.899	0.882	0.894	0.899	0.920	0.673
Fiber graph, 128 features	0.877	0.855	0.874	0.877	0.903	0.644
Axis graph, 128 features	0.743	0.654	0.748	0.751	0.818	0.488
Axis graph, 256 features	0.729	0.625	0.748	0.738	0.806	0.464
Hybrid (corrects PPCA), axis graph, 128 features	0.635	0.563	0.638	0.638	0.701	0.438
Hybrid, axis graph, 256 features, 2 convolutions	0.623	0.543	0.620	0.632	0.699	0.432
PPCA alone (same run)	0.685	0.638	0.721	0.662	0.717	0.639
PPCA + SIREN (same run)	0.572	0.408	0.516	0.642	0.720	0.639

Table A9: Noise models of the PPCA (earlier protocol: 30 scans, one seed, 11 measures; 4-mm limit; overall score).

	Isotropic		Per measure		Per cell	
q	alone	+ SIREN	alone	+ SIREN	alone	+ SIREN
5	0.729	0.578	0.730	0.579	0.735	0.584
10	0.705	0.566	0.707	0.568	0.710	0.572
20	0.688	0.561	0.689	0.561	0.692	0.563
40	0.667	0.553	0.668	0.554	0.675	0.558
80	0.648	0.546	0.647	0.546	0.653	0.552

Table A10: Per measure with the free-water tensor measures (11 measures, 3 seeds; four patterns).

Setting	FA	MD	RD	AD	FW-FA	FW-MD	FW-RD	FW-AD	FWF	NDI	ODI
PPCA alone	0.784	0.660	0.656	0.756	0.682	0.536	0.514	0.725	0.804	0.642	0.814
PPCA + SIREN	0.661	0.536	0.534	0.637	0.576	0.429	0.419	0.596	0.685	0.525	0.704
PPCA + SIREN, U-Net beyond	0.654	0.538	0.533	0.637	0.571	0.429	0.418	0.589	0.687	0.524	0.696

Table A11: Pooling HBCD and continuous age norms on 11 measures (3 seeds × 54 BCP scans; BCP scans in BCP units against the BCP-alone age-bin mean).

Setting (11 measures, phase-3b protocol)	Overall (SD)	0–3 mo	4–9 mo	10–60 mo
BCP alone, age bins	0.573 (0.007)	0.563	0.567	0.589
BCP alone, kernel age norm	0.584 (0.017)	0.598	0.562	0.593
Pooled tensor+NODDI, age bins	0.564 (0.010)	0.546	0.571	0.577
Pooled tensor+NODDI, kernel age norm	0.620 (0.052)	0.709	0.567	0.584
Pooled all measures, age bins	0.584 (0.010)	0.549	0.599	0.605
Pooled all measures, kernel age norm	0.589 (0.016)	0.561	0.596	0.611
Age norm alone: age-bin mean	1.000 (0.000)	1.000	1.000	1.000
Age norm alone: kernel	0.912 (0.020)	0.868	0.935	0.933
Age norm alone: spline (6/3 df)	0.913 (0.022)	0.869	0.933	0.937

Table A12: Residual cohort difference after the batch correction: median over cells of the difference of the cohort means (HBCD − BCP) in BCP SDs, per age window.

Age window (mo)	n BCP / HBCD	mean age BCP / HBCD	FA	MD	RD	AD	NDI	ODI	FWF
0-1.5	8 / 1213	0.4 / 0.9	+0.34	-0.09	-0.07	-0.16	+0.16	-0.29	+0.26
1.5-3.5	20 / 271	2.7 / 2.0	-0.84	+2.12	+2.08	+1.10	-0.68	+0.12	+1.72
3.5-6.5	60 / 353	5.1 / 5.3	+0.06	-0.28	-0.24	-0.23	+0.32	-0.07	+0.06
6.5-9.5	57 / 310	8.1 / 7.7	+0.01	+0.06	+0.01	+0.11	-0.11	-0.06	-0.06
9.5-16	110 / 145	12.5 / 12.1	+0.02	-0.16	-0.17	-0.03	-0.11	+0.07	-0.18

Table A13: BCP scans per acquisition label and age bin.

Acquisition	0–3 mo	4–9 mo	10–60 mo
`dir79select`	19	101	191
`shells06`	0	0	4
`shells06select`	9	16	83

Table A14: Tuning budget per model family: screened runs and validation trainings with their GPU time, from the training logs (one training per fold).

Family	Screen runs	Screen GPU h	Validation trainings	Validation GPU h
Functa	187	58.8	0	0.0
Partial VAE	8	1.2	0	0.0
Age-dependent PPCA	3	0.0	0	0.0
Graph U-Net	1	0.1	222	16.9
Transformer	12	2.9	68	27.3

Table A15: Blend weight tables refitted from the validation dumps (identical to `ENSEMBLE_TABLES` of DTIPlayground 0.9.8).

		SIREN cells (mm)					Fallback cells (mm)				
Table	Measure	0-0.5	0.5-1	1-2	2-3	3-4	4-6	6-10	10-20	>20	Family
BCP	FA	0.27	0.31	0.40	0.51	0.61	0.62	0.65	0.62	0.75	0.98
	MD	0.19	0.24	0.36	0.51	0.60	0.58	0.61	0.70	0.79	0.97
	RD	0.23	0.26	0.36	0.50	0.59	0.57	0.61	0.66	0.85	0.98
	AD	0.23	0.31	0.43	0.56	0.68	0.65	0.71	0.71	0.73	0.96
	NDI	0.16	0.24	0.36	0.49	0.61	0.70	0.68	0.70	0.87	0.82
	ODI	0.25	0.31	0.46	0.58	0.67	0.68	0.70	0.66	0.80	0.96
	FWF	0.20	0.26	0.41	0.47	0.64	0.72	0.70	0.73	0.87	0.93
BCP+HBCD	FA	0.30	0.44	0.57	0.67	0.71	0.69	0.72	0.66	0.69	0.91
	MD	0.31	0.36	0.48	0.62	0.67	0.64	0.61	0.61	0.59	0.87
	RD	0.32	0.38	0.50	0.64	0.69	0.68	0.67	0.64	0.65	0.90
	AD	0.34	0.46	0.57	0.68	0.74	0.68	0.68	0.65	0.59	0.86
	NDI	0.25	0.38	0.51	0.65	0.72	0.69	0.65	0.63	0.65	0.58
	ODI	0.34	0.47	0.61	0.72	0.75	0.71	0.75	0.68	0.66	0.81
	FWF	0.31	0.37	0.46	0.55	0.66	0.56	0.54	0.52	0.59	0.63

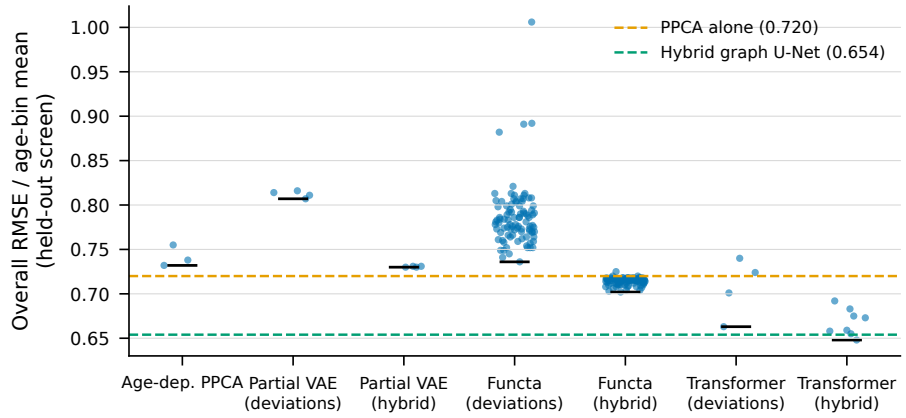


Figure A1: Quick held-out screen: overall error of every screened configuration (dots; bar: best) with PPCA alone and the hybrid graph U-Net on the same split.

Table A16: O1 and O3: norm and PPCA fitted on a share of the training subjects (learning curve) or with 7 finer age bins, scored on the hidden values of the validation (relative to the 3-bin age-bin mean; paired whole-profile ratio against the full training set with the 95% subject-bootstrap CI). Seeds completed: 3.

Norm and PPCA from	Method	Training scans	Overall	Gap	End	Whole	Region	Whole: paired vs 100%
25% of training subjects	PPCA	99	0.898	0.817	0.976	0.878	0.919	1.167 [1.140, 1.196]
25% of training subjects	PPCA + kNN	99	0.835	0.689	0.865	0.857	0.927	1.187 [1.151, 1.226]
50% of training subjects	PPCA	198	0.793	0.738	0.854	0.769	0.812	1.067 [1.052, 1.083]
50% of training subjects	PPCA + kNN	198	0.719	0.610	0.724	0.734	0.806	1.073 [1.056, 1.094]
75% of training subjects	PPCA	295	0.754	0.704	0.806	0.729	0.775	1.021 [1.013, 1.029]
75% of training subjects	PPCA + kNN	295	0.681	0.583	0.683	0.691	0.768	1.022 [1.013, 1.031]
100% of training subjects	PPCA	395	0.733	0.684	0.775	0.712	0.760	–
100% of training subjects	PPCA + kNN	395	0.664	0.567	0.659	0.675	0.753	–
7 finer age bins	norm alone	395	0.951	0.937	0.968	0.946	0.954	–
7 finer age bins	PPCA	395	0.763	0.713	0.811	0.740	0.788	1.029 [1.015, 1.043]
7 finer age bins	PPCA + kNN	395	0.694	0.588	0.696	0.708	0.786	1.035 [1.020, 1.051]

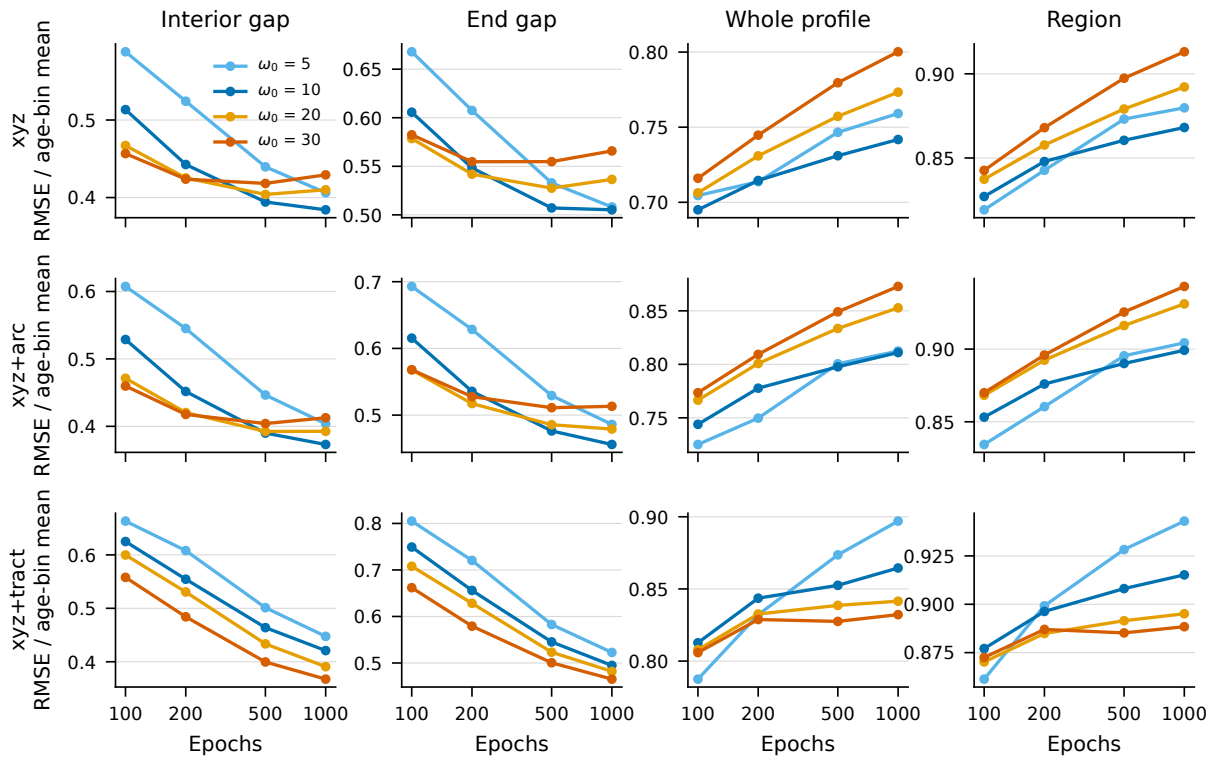


Figure A2: SIREN on the deviations from the age-bin norm: error against epochs per input (rows), ω_0 (colours) and pattern (columns). Earlier protocol: 30 scans, one seed, 11 measures.

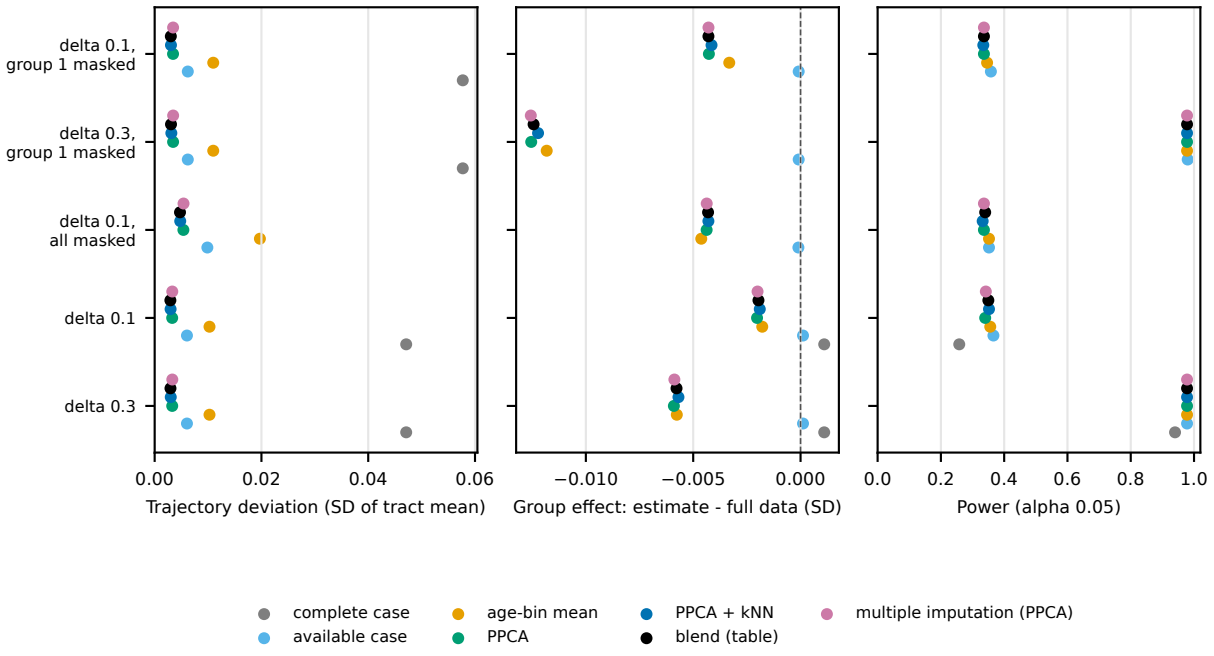


Figure A3: E4: trajectory deviation from the complete-data curve, group-effect estimate minus the complete-data estimate (between-scan SD), and power per imputer and scenario.

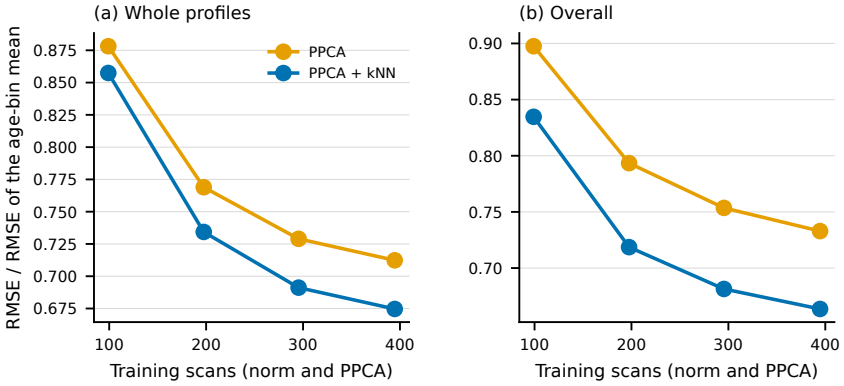


Figure A4: O1: error of PPCA and PPCA + kNN against the number of normative training scans (norm and PPCA fitted on the same subset).

Table A17: Quick held-out screen (one split, 30 test subjects, 2 rounds; 7 measures): best result per family and target, selected on the test split itself; "Runs": configurations and test-time variants screened. Not comparable with Table 3.

Model	Target	Runs	Overall	Gap	End	Whole	Region	Tensor	NODDI
PPCA ($q = 80$)	–	1	0.720	0.665	0.780	0.702	0.732	0.683	0.774
Age-dependent PPCA	–	3	0.732	0.685	0.792	0.712	0.737	0.704	0.783
Partial VAE	deviations	4	0.807	0.783	0.844	0.798	0.801	0.803	0.885
Partial VAE	PPCA correction	4	0.730	0.673	0.789	0.712	0.745	0.696	0.786
Functa	deviations	93	0.736	0.674	0.752	0.730	0.787	0.692	0.836
Functa	PPCA correction	94	0.702	0.641	0.750	0.689	0.728	0.671	0.761
Graph U-Net	PPCA correction	1	0.654	0.570	0.678	0.658	0.711	0.483	0.672
Transformer	deviations	4	0.663	0.556	0.641	0.718	0.737	0.452	0.658
Transformer	PPCA correction	8	0.648	0.537	0.640	0.697	0.720	0.433	0.639

Table A18: Error relative to the age-bin mean per measure (four patterns, mean over age bins and seeds), and for a whole missing family. ‡First row: absolute RMSE of the age-bin mean (FA, NDI, ODI, FWF $\times 10^3$; MD, RD, AD $\times 10^6$ mm^2/s).

Setting	FA	MD	RD	AD	NDI	ODI	FWF
Age-bin mean: absolute RMSE‡	46.3	102.6	101.6	125.8	65.5	51.8	56.2
Phase 2: SIREN on deviations	0.688	0.638	0.628	0.714	0.613	0.768	0.769
PPCA alone	0.784	0.657	0.652	0.756	0.659	0.818	0.808
PPCA + SIREN	0.663	0.533	0.533	0.639	0.535	0.705	0.689
U-Net alone	0.698	0.604	0.599	0.695	0.605	0.760	0.753
Transformer alone (400)	0.668	0.572	0.570	0.648	0.563	0.699	0.705
Released default (table blend)	0.610	0.504	0.504	0.587	0.505	0.645	0.631
With HBCD reference (table blend)	0.578	0.480	0.478	0.559	0.481	0.610	0.594
Whole family missing (tensor: FA–AD; NODDI: NDI–FWF)							
PPCA alone	0.757	0.648	0.639	0.741	0.758	0.832	0.803
U-Net alone	0.523	0.512	0.502	0.585	0.729	0.680	0.700
Transformer alone (400)	0.490	0.374	0.379	0.456	0.681	0.616	0.573
Released default (table blend)	0.490	0.374	0.379	0.455	0.678	0.615	0.574

Table A19: Absolute RMSE (four patterns; mean over patterns and seeds) per age bin; FA, NDI, ODI, FWF $\times 10^3$, MD, RD, AD $\times 10^6\,\mathrm{mm}^2/\mathrm{s}$. Released default and HBCD-reference blend cross-fitted. All main settings, pooled over age bins: Appendix Table A4.

Age bin	Setting	FA	MD	RD	AD	NDI	ODI	FWF
0–3 mo	Age-bin mean	46.0	121.9	122.5	138.6	68.1	66.1	63.0
	Released default	26.0	57.9	56.2	80.9	37.0	40.1	41.1
	With HBCD reference	24.2	51.9	50.0	74.1	33.5	36.1	37.1
4–9 mo	Age-bin mean	40.5	91.3	87.5	117.7	58.8	42.0	51.1
	Released default	26.6	44.6	44.2	65.6	30.4	28.1	31.0
	With HBCD reference	25.4	44.3	43.7	64.2	29.8	27.3	30.1
10–60 mo	Age-bin mean	52.4	94.5	94.8	121.1	69.7	47.3	54.5
	Released default	31.4	52.0	51.7	75.3	32.0	30.1	35.0
	With HBCD reference	30.0	50.2	49.7	72.5	31.4	29.0	33.4

Table A20: Selected paired contrasts, four patterns, 162 scan × seed pairs (96 subjects): geometric mean ratio of per-scan errors (compared / reference), 95% CIs over pairs and from the subject-level bootstrap, Holm-adjusted Wilcoxon p (pairs / subject means), and the share of pairs where the compared setting is better.

Contrast (compared vs reference)	Ratio	95% CI (pairs)	95% CI (subjects)	p_{Holm} (pairs / subjects)	Better
Population model					
PPCA + SIREN vs PPCA alone	0.834	0.827–0.841	0.827–0.842	$< 10^{-26}$ / $< 10^{-15}$	100%
PPCA + kNN vs PPCA + SIREN	1.082	1.075–1.088	1.074–1.089	$< 10^{-26}$ / $< 10^{-15}$	1%
$q = 40$ vs $q = 80$ (+ SIREN)	1.008	1.002–1.014	1.002–1.014	0.036 / 0.008	42%
$q = 120$ vs $q = 80$ (+ SIREN)	1.010	1.003–1.017	1.002–1.017	$< 10^{-4}$ / 4.3×10^{-4}	36%
Graph U-Net					
U-Net alone vs PPCA + SIREN	1.085	1.075–1.095	1.074–1.096	$< 10^{-23}$ / $< 10^{-14}$	7%
U-Net beyond 4 mm vs PPCA beyond	0.993	0.990–0.995	0.990–0.996	$< 10^{-7}$ / $< 10^{-5}$	70%
U-Net blend 0.25/0.75 vs PPCA + SIREN	0.966	0.963–0.970	0.962–0.970	$< 10^{-25}$ / $< 10^{-15}$	95%
Transformer					
Transformer alone vs PPCA + SIREN	1.059	1.050–1.069	1.050–1.069	$< 10^{-20}$ / $< 10^{-14}$	13%
Transformer alone vs U-Net alone	0.976	0.970–0.983	0.970–0.983	$< 10^{-9}$ / $< 10^{-6}$	74%
Transformer blend 0.25/0.75 vs PPCA + SIREN	0.947	0.944–0.950	0.943–0.950	$< 10^{-26}$ / $< 10^{-15}$	100%
Transformer blend vs U-Net blend (0.25/0.75)	0.980	0.978–0.982	0.977–0.982	$< 10^{-24}$ / $< 10^{-15}$	94%
Blending					
Precision weights vs fixed 0.25/0.75	1.012	1.010–1.015	1.010–1.015	$< 10^{-12}$ / $< 10^{-9}$	24%
400 vs 200 epochs (fixed blend)	0.988	0.986–0.990	0.986–0.990	$< 10^{-22}$ / $< 10^{-12}$	88%
Weight table vs fixed (400 epochs)	0.987	0.984–0.989	0.984–0.989	$< 10^{-19}$ / $< 10^{-11}$	87%
Released default vs PPCA + SIREN	0.923	0.918–0.928	0.917–0.928	$< 10^{-26}$ / $< 10^{-15}$	99%
Not helpful					
Acquisition correction vs none	1.001	0.999–1.004	0.999–1.004	0.45 / 0.15	48%
Second cohort					
Pooled PPCA + SIREN vs BCP alone	1.029	1.015–1.043	1.014–1.044	$< 10^{-5}$ / $< 10^{-5}$	31%
Reference transformer vs BCP transformer (alone)	0.923	0.913–0.933	0.911–0.934	$< 10^{-22}$ / $< 10^{-13}$	93%
Reference vs BCP transformer (table blend)	0.952	0.945–0.959	0.943–0.960	$< 10^{-22}$ / $< 10^{-12}$	92%

Table A21: Pooling per age bin (paired, compared / reference, 95% subject-bootstrap CI; seeds 0–2, BCP scans).

Contrast (compared vs reference)	All	0–3 mo	4–9 mo	10–60 mo
Pooled PPCA + SIREN vs BCP PPCA + SIREN	1.029 [1.014, 1.044]	0.977 [0.939, 1.011]	1.057 [1.035, 1.082]	1.039 [1.025, 1.052]
Pooled fixed blend vs released default	0.998 [0.985, 1.010]	0.957 [0.923, 0.986]	1.024 [1.008, 1.043]	1.002 [0.994, 1.010]
Pooled fixed blend vs HBCD-reference fixed blend	1.016 [1.008, 1.024]	0.993 [0.971, 1.011]	1.029 [1.018, 1.042]	1.019 [1.013, 1.026]
HBCD-reference transformer vs BCP transformer (both 200 ep.)	0.889 [0.877, 0.900]	0.865 [0.833, 0.895]	0.905 [0.891, 0.922]	0.889 [0.878, 0.900]
HBCD-reference fixed blend vs BCP fixed blend (both 200 ep.)	0.957 [0.952, 0.963]	0.938 [0.923, 0.952]	0.969 [0.962, 0.977]	0.960 [0.955, 0.965]
Mean of BCP and reference transformers vs BCP (fixed, 400 ep.)	0.974 [0.970, 0.977]	0.957 [0.948, 0.965]	0.981 [0.978, 0.985]	0.978 [0.975, 0.981]

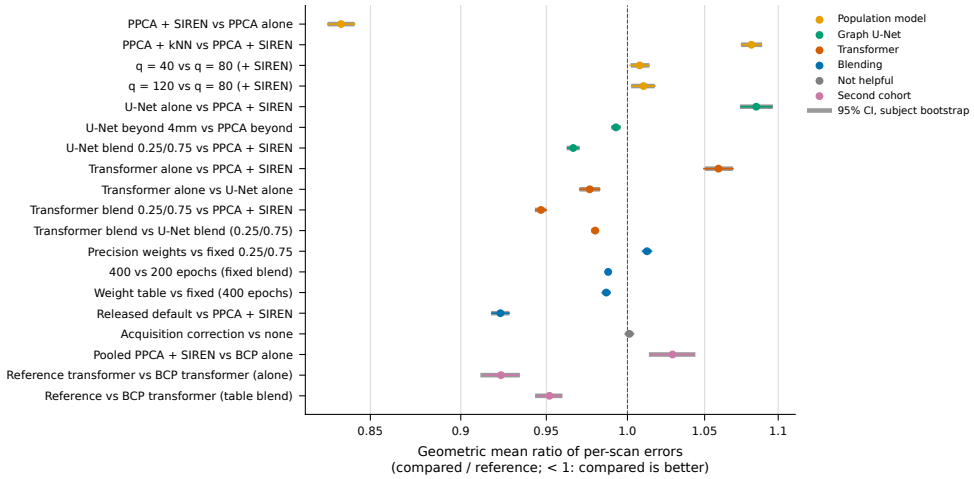


Figure A5: Paired contrasts of Table A20: geometric mean ratio (dot), 95% CI over pairs (coloured line) and subject-level bootstrap CI (grey bar).

Table A22: E2: real-mask pattern with all donors, all settings of the validation runs (BCP: E1 and the BCP-alone run of E5; HBCD: E3).

Setting	BCP, untouched subjects (E1)	BCP, seeds 0–2 (E5i)	HBCD (E3)
age-bin mean	1.000	–	1.000
PPCA alone	0.801	0.751	1.097
PPCA + kNN	0.758	0.742	1.083
PPCA + SIREN	0.777	0.753	1.086
Transformer alone	0.757	–	0.767
Fixed blend (SIREN base)	0.723	–	0.816
Fixed blend (kNN base)	0.718	–	0.818
Precision blend	0.724	–	0.901
Shipped table (SIREN base; released default)	0.713	–	0.799
Shipped table (kNN base)	0.716	–	0.805

Table A23: Real-mask pattern by donor load (tracts missing in the donor; all donors), with the share of the real-mask cells; error relative to the age-bin mean.

Cohort	Donor missing tracts	Cells (%)	PPCA + *k*NN	PPCA + SIREN	Transformer alone	Shipped table (released default)	Table refitted on the cohort (cross-fitted)
BCP (E1)	1	19	0.746	0.753	0.754	0.693	0.694
	2–5	30	0.756	0.768	0.738	0.690	0.690
	6–15	17	0.794	0.813	0.787	0.746	0.746
	16–34	23	0.677	0.712	0.694	0.656	0.656
	≥35	10	0.827	0.833	0.765	0.759	0.762
HBCD (E3)	1	2	1.217	1.201	0.853	0.891	0.840
	2–5	15	0.969	0.972	0.775	0.794	0.768
	6–15	28	0.915	0.919	0.699	0.721	0.690
	16–34	19	0.916	0.922	0.751	0.762	0.742
	≥35	36	1.449	1.454	0.829	0.884	0.829